Journal of Sports Sciences

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/rjsp20

The efficacy of exercise interventions for all types of inpatients across mental health settings: A systematic review and meta-analysis of 47 studies

Rebecca Martland, Nicole Korman, Joseph Firth & Brendon Stubbs

To cite this article: Rebecca Martland, Nicole Korman, Joseph Firth & Brendon Stubbs (2023) The efficacy of exercise interventions for all types of inpatients across mental health settings: A systematic review and meta-analysis of 47 studies, Journal of Sports Sciences, 41:3, 232-271, DOI: 10.1080/02640414.2023.2207855

To link to this article: https://doi.org/10.1080/02640414.2023.2207855

© 2023 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

View supplementary material

Published online: 03 May 2023.

Submit your article to this journal

Article views: 5666

View related articles

View Crossmark data

Citing articles: 7 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=rjsp20

JOURNAL OF SPORTS SCIENCES
2023, VOL. 41, NO. 3, 232–271
https://doi.org/10.1080/02640414.2023.2207855

Routledge
Taylor & Francis Group

PHYSICAL ACTIVITY, HEALTH AND EXERCISE

OPEN ACCESS Check for updates

The efficacy of exercise interventions for all types of inpatients across mental health settings: A systematic review and meta-analysis of 47 studies

Rebecca Martland [a], Nicole Korman [b,c], Joseph Firth [d,e] and Brendon Stubbs [a,f]

[a]King's College London, Institute of Psychiatry, Psychology and Neuroscience (IoPPN), London, UK; [b]Metro South Addiction and Mental Health Services, Brisbane, Australia; [c]School of Medicine, University of Queensland, Brisbane, Australia; [d]Western Sydney University, Westmead, NSW, NICM Health Research Institute, Australia; [e]Faculty of Biology, Medicine & Health, Division of Psychology and Mental Health, University of Manchester, Manchester, United Kingdom; [f]South London and Maudsley NHS Foundation Trust, London, United Kingdom

ABSTRACT

This systematic review and meta-analysis investigated the benefits, safety and adherence of exercise interventions delivered in inpatient mental health settings, quantified the number of exercise trials that provided support to maintain engagement in exercise post-discharge, and reported patient feedback towards exercise interventions. Major databases were searched from inception to 22.06.2022 for intervention studies investigating exercise in mental health inpatient settings. Study quality was assessed using Cochrane and ROBINS-1 checklists. Fifty-six papers were included from 47 trials (including 34 RCTs), bias was high. Exercise improved depression (Standardised mean difference = −0.416; 95% Confidence interval −0.787 to −0.045, $N = 15$) compared to non-exercise comparators amongst people with a range of mental illnesses, with further (albeit limited) evidence suggesting a role of exercise in cardiorespiratory fitness and various other physical health parameters and ameliorating psychiatric symptoms. No serious exercise-related adverse events were noted, attendance was ≥80% in most trials, and exercise was perceived as enjoyable and useful. Five trials offered patients post-discharge support to continue exercise, with varying success. In conclusion, exercise interventions may have therapeutic benefits in inpatient mental health settings. More high-quality trials are needed to determine optimal parameters, and future research should investigate systems to support patients to maintain exercise engagement once discharged.

ARTICLE HISTORY
Received 27 May 2021
Accepted 18 April 2023

KEYWORDS
Exercise; systematic review; mental health; depression; hospital

Introduction

Patients with severe mental illnesses (SMI), including schizophrenia, bipolar disorder and major depressive disorder (MDD), have a 78% higher risk for developing cardiovascular disease (CVD) compared to healthy controls (Correll et al., 2017). Moreover, the risk of developing coronary heart disease is 26–41% higher in anxiety/stress disorders versus the general population (Emdin et al., 2016; Roest et al., 2010). Further, the occurrence of diabetes and metabolic syndrome is two-fold higher across those with SMI, anxiety/stress disorders, and alcohol and substance use disorders (AUD; SUD) (Correll et al., 2017; De Hert et al., 2009) (Whiteford et al., 2013). This increased risk of physical co-morbidity contributes to 10–25 years shortened life expectancy for SMI (Correll et al., 2015) and 7–20 years shortened life expectancy for AUDs, SUDs, anxiety and stress disorders (Denollet et al., 2009; Edward et al., 2014; Lohr et al., 2015) (Weitoft & Rosén, 2005).

Structured exercise can improve cardiorespiratory and cardiometabolic parameters in people with mental illnesses (Firth et al., 2015) (Vancampfort, Rosenbaum, et al., 2015). Exercise has benefits for cognitive functioning, depression severity, sleep quality, alcohol intake and psychiatric symptoms (Czosnek et al., 2019; Stubbs et al., 2018; Smith & Lynch, 2012). Despite the breadth in the evidence basis in terms of exercise modality, intervention duration, frequency and mental health diagnosis, the majority of exercise intervention studies have recruited patients with mental illnesses receiving care from community and outpatient settings. Less research has been conducted in inpatient settings, where patients are more acutely unwell, receive higher doses of psychotropic medication (Barbui et al., 2007) and polypharmacy (Weinmann et al., 2004).

People receiving psychiatric inpatient care show significantly greater levels of sedentary behaviour, reduced levels of exercise, and substantially reduced objectively measured moderate/vigorous physical activity levels than healthy controls (Kruisdijk et al., 2017). Reasons for inactivity and sedentary behaviour of people receiving inpatient care are likely to be multifactorial and associated with impact of acutely distressing symptoms, sedative effects of neuroleptics and reduced opportunity for physical activity due to lack of suitable facilities and restrictions on leave to access physical activity for those detained under mental health legislation (Firth et al., 2016; Gilburt et al., 2008). Moreover, people receiving psychiatric inpatient treatment may struggle to manage their physical health whilst acutely unwell and commonly experience acute mental health symptoms that may not fully remit with

CONTACT Rebecca Martland rebecca.martland@kcl.ac.uk Department of Psychosis Studies, Institute of Psychiatry, Psychology and Neuroscience (IoPPN), King's College London, London, United Kingdom

This article has been republished with minor changes. These changes do not impact the academic content of the article.
Supplemental data for this article can be accessed online https://doi.org/10.1080/02640414.2023.2207855.

© 2023 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

traditional first-line treatments (i.e., medication or therapies) alone (Kennedy et al., 2014). Inpatient exercise programmes may provide crucial opportunities to impact on both the physical and mental recovery of acutely distressed inpatients.

One 2014 review examined the effect of exercise interventions in people hospitalized with SMI and anxiety disorders (Stanton & Happell, 2014a), observing reductions in depression and improvements in quality-of-life following exercise interventions in those hospitalized with depression. There was preliminary evidence for improvements in psychiatric symptoms and aerobic fitness in inpatients with schizophrenia, bipolar disorder, and anxiety disorders. However, the review included only eight studies of low-to-moderate quality thus limiting generalizability (Stanton & Happell, 2014a). Moreover, a 2018 meta-analysis examined the anti-depressant effects of aerobic exercise in MDD and found large therapeutic effects (Morres et al., 2019). Since this time, many more inpatient exercise studies have been conducted, although there is a lack of clarity on the therapeutic effects of exercise across all mental health disorders, and post-discharge follow-up.

The aim of this systematic review and meta-analysis was to investigate the benefits, safety, adherence and description of exercise interventions for individuals receiving inpatient mental health treatment, using evidence from controlled and uncontrolled intervention studies. Secondary aims were to quantify and illustrate the form of any support to maintain engagement in exercise interventions once discharged from hospital. Moreover, we report patient feedback to the included exercise regimes, and whether exercise impacted length of hospital stay.

Methods

The systematic review was reported according to PRISMA guidelines following a pre-determined protocol (Prospero CRD42020183359).

Searches and study selection

Medline, PsychoInfo and Embase were searched from inception to 22.06.2022 for intervention studies investigating any form of exercise among people receiving inpatient treatment for a SMI (including schizophrenia spectrum disorders, bipolar disorder, MDD); anxiety and stress disorders; AUD and SUD's; and eating disorders. All study designs were included (including RCTs, non-randomized controlled trials (NRCTs), pre- and post-test studies, naturalistic trials).

The search terms used are found in Supplementary Material 1. The reference lists of included articles were hand-searched.

Definition of exercise interventions

Aerobic exercise (walking, running, cycling or similar continuous activities) and/or non-aerobic exercise (free weight training, resistance training, body weight training) were included. Yoga, tai-chi and dance were excluded seen as these activities may benefit mental health through additional factors distinct from the physical activity itself (Stubbs et al., 2018).

Eligibility criteria

Two authors independently assessed articles based on the following criteria: 1) Intervention study investigating exercise, in a mental health inpatient setting, in any age. Where a control group was used, we considered any form of control group including no treatment, therapeutic and lifestyle interventions and pharmacotherapy; 2) Exercise interventions of any duration and frequency, excluding single exercise session studies; 3) a study population which included any forms of mental illness.

Non-English language articles, conference abstracts and dissertation theses were excluded. Exercise interventions that included therapeutic and lifestyle components and studies conducted in supported housing/residential facilities were excluded.

Outcomes

Primary outcomes

Changes in mental health parameters including psychiatric symptom scores, depression, anxiety, sleep quality and substance/alcohol cravings.

Secondary outcomes

Changes in cardiometabolic risk factors (e.g., BMI, systolic and diastolic blood pressure (SBP, DBP), heart rate (HR), lipid profiles), cardiorespiratory fitness (e.g., VO2max/VO2peak) and cognitive functioning. In the review planning stage (Prospero CRD42020183359) the decision was undertaken to report a wider range of outcomes including appetite/cravings, medication dose and sedentary behaviour. These outcomes have not been reported due to paucity of trials for each of these outcomes.

Further, we reported: 1) adverse events (AEs); 2) completion and drop-out rates; 3) referral to community exercise and support to exercise post discharge; 4) participant feedback/enjoyment; 5) length of inpatient stay.

Search results and data extraction

The title and abstract of all studies identified were reviewed, and relevant full-texts reviewed to determine eligibility. Eligibility was determined by three independent researchers (RM, BS, NK).

Data was extracted, from the papers selected for inclusion in the review, by one researcher (RM) and verified by a second researcher (BS). For each intervention study, we reported: study design, sample size, participant demographics, intervention descriptions (including control interventions), effect sizes (ES), adherence; and AEs.

Quality assessment

The quality of the included RCTs were assessed using The Cochrane Collaboration's tool for assessing risk of bias in randomized trials and The Risk of Bias in Non-Randomized Studies of Interventions (ROBINS-I) for non-RCTs (J. A. C. Sterne et al., 2019; J. A. Sterne et al., 2016). Assessment was carried out by two independent researchers (RM and NK).

For each study, the intervention description was reported using the Consensus on Exercise Reporting Template (CERT) (Slade et al., 2016) (Supplementary Material 2).

Data synthesis

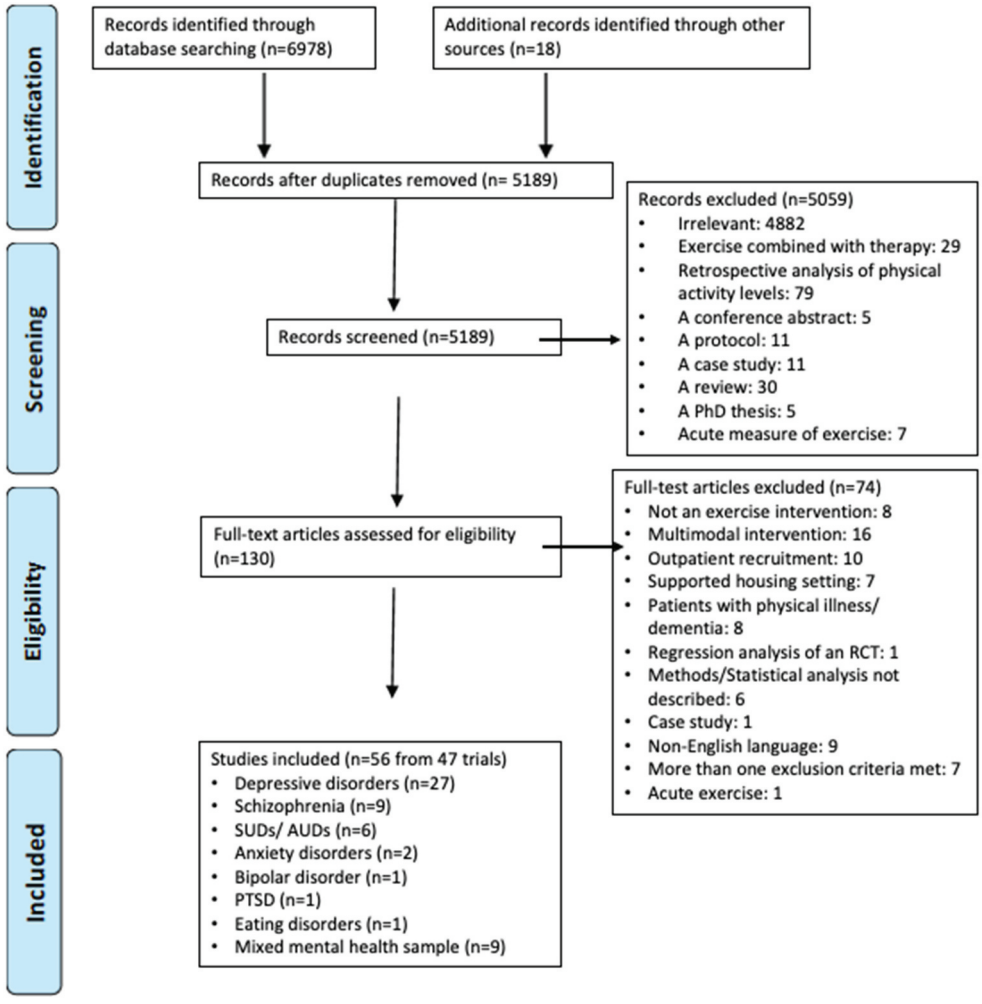
Primary and secondary outcomes were summarised narratively. We subdivided intervention studies based on mental health diagnosis and type of trial (RCTs, NRCTs, pre-post-test) and narratively summarised the number of studies that supported a significant improvement ($p \leq 0.05$) in each outcome.

Additionally, we conducted between group meta-analyses for RCTs comparing exercise with non-exercise comparators for depression and BMI using data from all RCTs across all mental health diagnoses. Due to variation in HIIT protocols and participant characteristics, we used a random-effects model calculating the standardised mean difference (SMD) and 95% confidence intervals (95% CI) using the difference between the two groups' pre-post change scores. The mean and standard deviation (SD) of pre-post scores was extracted for each study to calculate effect sizes. Heterogeneity was assessed using the I^2 statistic. Publication bias was assessed with the Begg-Mazumdar Kendall's tau. Comprehensive Meta-Analysis Software (CMA) was used for analysis. Meta-analyses could not be conducted for other primary and secondary outcomes due to heterogeneity/paucity of studies for each outcome and mental health diagnosis.

Results

Search results and included studies

Search results are in Figure 1. Overall, 56 papers were included (from 47 trials) encompassing exercise interventions in depressive disorders ($N = 27$ from 20 trials, $n = 892$ participants including control groups) (Imboden et al., 2020; Kerling et al., 2017), schizophrenia spectrum disorders ($N = 9$ from 8 trials, $n = 365$) (Mazyarkin et al., 2019; Shimada et al., 2019), bipolar disorder ($N = 1$, $n = 49$) (Ng et al., 2007), SUDs and/or AUDs ($N = 6$, $n = 207$) (Schneider et al., 2016; McCartney et al., 2021), post-traumatic stress disorder (PTSD) ($N = 1$, $n = 81$) (Rosenbaum et al., 2015), anxiety ($N = 2$, $n = 122$) (Gaul-Aláčová et al., 2005;



Key terms: AUD= Alcohol use disorder, RCT= randomized controlled trial, PTSD= post-traumatic stress disorder, SUD= substance use disorder

Figure 1. Prisma flow chart. Note: Keyterms: AUD = Alcohol use disorder, RCT = randomized controlled trial, PTSD = post-traumatic stress disorder, SUD = substance use disorder. bstance use disorder

Martinsen, Hoffart, & Solberg, 1989), anorexia nervosa ($N = 1$, $n = 41$) (Kern et al., 2020), and inpatient samples with a range of mental health disorders ($N = 9$ from 8 trials, $n = 591$) (Kim et al., 2014; Philippot et al., 2022). Mean age of participants ranged from 15.5 to 68.7 years.

Thirty-four trials (across 42 papers) followed an RCT design (Imboden et al., 2020) (Haussleiter et al., 2020; Imboden et al., 2021) (Jae et al., 2014; Kerling et al., 2015, 2017; Manjunath et al., 2013; Roy et al., 2018) (Gary & Guthrie, 1972; Palmer et al., 1995; Schneider et al., 2016; Shimada et al., 2019) (Kim et al., 2014; Knapen et al., 2003, 2005; Lan et al., 2022; Martinsen, Hoffart, & Solberg, 1989; Philippot et al., 2022; Rosenbaum et al., 2015; Sexton et al., 1989). Other study designs included a semi-randomized controlled trial ($N = 1$ across 2 papers) (Wunram et al., 2018, 2021), NRCT ($N = 5$) (Dürmüş et al., 2020; Gaul-Aláčová et al., 2005; Heggelund et al., 2011; Mazyarkin et al., 2019; Song et al., 2018), a pre-post-test-controlled trial ($N = 1$) (Kerling et al., 2016), pre-post-test ($N = 5$) (Kern et al., 2020; Martinsen, Sandvik, et al., 1989; Stanley et al., 2019; Tsukue & Shohoji, 1981; Woodward et al., 2018) and a retrospective cohort study ($N = 1$) (Ng et al., 2007). Where a control group was utilised, activities included treatment as usual (TAU) ($N = 18$) (Kerling et al., 2018) (Bosscher, 1993; Haussleiter et al., 2020; Legrand & Neff, 2016; Roy et al., 2018; Schuch et al., 2014, 2015; Wunram et al., 2021) (Kerling et al., 2015, 2017) (Jae et al., 2014; Kurebayashi et al., 2022) (Dürmüş et al., 2020; Gaul-Aláčová et al., 2005; Kim et al., 2014; Lan et al., 2022; Ng et al., 2007; Rosenbaum et al., 2015), stretching/relaxation ($N = 10$) (Gerber et al., 2020; Imboden et al., 2020; Legrand & Neff, 2016) (CWH et al., 2014; Knapen et al., 2003, 2005; Knubben et al., 2007; McCartney et al., 2021; Philippot et al., 2022), yoga ($N = 1$) (Manjunath et al., 2013), occupational therapy (OT) ($N = 1$) (Buschert et al., 2019), computer games ($N = 1$) (Heggelund et al., 2011), association splitting ($N = 1$) (Schneider et al., 2016), passive muscular training ($N = 1$) (Wunram et al., 2018, 2021), and electroconvulsive therapy ($N = 1$) (Salehi et al., 2016).

Tables 1–4 respectively compile information on sample and intervention details, and primary and secondary outcomes.

Quality of included studies and exercise reporting

A Cochrane risk of bias assessment table and ROBINS-1 tables for non-RCTs are provided in Supplementary Materials 3 and 4, respectively.

Briefly, 26/42 RCTs scored as low risk of bias for random sequence generation, the remaining RCTs did not divulge into randomisation procedure, 18/42 RCTs scored low bias for allocation concealment, 19/42 RCTs scored low bias for blinding of outcome assessment and 26/42 RCTs scored low risk of bias for incomplete outcome data.

In terms of non-RCTs, one study was rated low bias, four studies as moderate bias and nine studies as high risk of bias.

Supervision was noted in 40 trials (85.1%), seven trials (Dürmüş et al., 2020; Gary & Guthrie, 1972; Gaul-Aláčová et al., 2005; Jae et al., 2014; Schneider et al., 2016; Song et al., 2018; Stanley et al., 2019) did not provide supervision details, and no trial reported unsupervised exercise. Exercise was mainly supervised by an exercise instructor, physiotherapist, occupational therapist, a registered nurse or a researcher (Knubben et al., 2007; Kurebayashi et al., 2022; Legrand & Neff, 2016; Salehi et al., 2016; Schuch et al., 2014, 2015; Shachar-Malach et al., 2015; Wunram et al., 2018) (Lan et al., 2022; McCartney et al., 2021; Palmer et al., 1995; Philippot et al., 2022; Shimada et al., 2020). Twenty trials conducted exercise in group settings (Buschert et al., 2019; Haussleiter et al., 2020; Heissel et al., 2015; Martinsen, Hoffart, & Ø, 1989; Wunram et al., 2021) (Manjunath et al., 2013; Mazyarkin et al., 2019; Wunram et al., 2018) (Martinsen, Hoffart, & Solberg, 1989; Ng et al., 2007; Palmer et al., 1995; SY et al., 2016; Tsukue & Shohoji, 1981) (Kim et al., 2014; Knapen et al., 2003, 2005; Lan et al., 2022; Philippot et al., 2022; Sexton et al., 1989), 9 trials conducted 1:1 sessions (Kerling et al., 2016; Knubben et al., 2007; Legrand & Neff, 2016; McCartney et al., 2021; Rosenbaum et al., 2015; Salehi et al., 2016; Schuch et al., 2014, 2015; Shachar-Malach et al., 2015; Woodward et al., 2018) and 16 trials did not report delivery method (Imboden et al., 2020) (CWH et al., 2014; Dürmüş et al., 2020; Gary & Guthrie, 1972; Gaul-Aláčová et al., 2005; Gerber et al., 2018; Hanssen, Minghetti, Faude, et al., 2018; Hanssen, Minghetti, Faude, Schmidt-Trucksäss, et al., 2018; Heggelund et al., 2011; Jae et al., 2014; Kahl et al., 2016; Kerling et al., 2015, 2017; Martinsen, Sandvik, et al., 1989; Minghetti et al., 2018; Schneider et al., 2016; Song et al., 2018; Stanley et al., 2019). In one trial, the first three sessions took part individually to establish optimal pace and subsequent sessions ran in groups (Bosscher, 1993), and in another trial a mixture of group and individual sessions were offered (Shimada et al., 2019, 2020). Forty trials tailored exercise intensity to each individual's bodyweight/cardiorespiratory fitness/movement limitations.

Meta-analysis

Exercise resulted in a moderate reduction in depression severity compared to non-exercise comparators with high heterogeneity (SMD: −0.416; 95%CI: −0.787 to −0.045; $I^2 = 79.690\%$; $N = 15$) (Imboden et al., 2020; Legrand & Neff, 2016) (CWH et al., 2014; Haussleiter et al., 2020; Knubben et al., 2007; Salehi et al., 2016)– (Kerling et al., 2015; Manjunath et al., 2013; Philippot et al., 2022; Rosenbaum et al., 2015; Roy et al., 2018). There was no evidence of publication bias.

Following exercise, no change in BMI was observed compared to non-exercise comparators (SMD: −0.212; 95%CI: −0.507 to −0.084; $I^2 = 0\%$; $N = 5$) (Imboden et al., 2020; Jae et al., 2014; Kahl et al., 2016; Kim et al., 2014; Rosenbaum et al., 2015).

Depressive disorders

Twenty exercise trials (across 27 papers) were conducted in inpatients with MDD (562 = exercise and 330 = control groups) including 18 RCTs (Imboden et al., 2020)– (Kerling et al., 2015, 2017; Roy et al., 2018)], one semi-randomised trial[(Wunram et al., 2018)] and one pre-post-test [(Kerling et al., 2016)].

Primary Outcomes

Seven trials (six RCTs (CWH et al., 2014; Haussleiter et al., 2020; Kerling et al., 2015; Legrand & Neff, 2016; Roy et al., 2018; Schuch et al., 2015) and one semi-randomised trial (Wunram et al., 2018)) measured depression following exercise and non-active control, of which 6/7 studies (85.7%) (CWH et al., 2014;

Table 1. Basic characteristics of included RCTs including study design, population characteristics and details of exercise and control interventions.

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/long stay)	Intervention	Control intervention
Lan et al., (2022)	Two-armed RCT. Mental health inpatients were randomised to 8 weeks of health exercises or regular care.	Exercise = 60; Control = 60	Inpatients aged 20–70 years old with schizophrenia, bipolar, organic psychosis and SUD. (50% male in each group, majority aged 41–60 years).	Long stay rehabilitation ward. 80% of participants had been hospitalised for 5+ years.	8 weeks of stretching, upper and lower body exercises twice weekly. Sessions totalled 50 min.	Regular care.
Philippot et al., (2022)	Two-armed RCT. Adolescent inpatients were randomised to 6 weeks of exercise or relaxation.	Exercise = 26; Control = 26	Inpatients aged 12–19 years old with depression or anxiety. (Exercise: 40% male, aged 15.5 ± 1.8 years; con: 35% male, aged 15.2 ± 1.5).	Acute: 6-week hospitalisation window.	6 weeks of aerobic group games and muscle strengthening at 40–59% HRR 3–4 times weekly (20 sessions). Each session totalled 1 hour.	6 weeks of board games and relaxation exercises without physical exertion (20 one-hour sessions).
Kurebayashi et al., 2021	Two-armed RCT. Inpatients with schizophrenia were randomised to 8 weeks of exercise or treatment as usual.	Exercise = 5; Control = 17	Inpatients aged 20+ with a DSM diagnosis of schizophrenia. (Exercise: age 50.3 ± 14 years, BMI 24.9 ± 4.4; Con: 59.7 ± 13 years, BMI 23.7 ± 2.3).	Long stay: average length of hospital stay was 193.5 ± 183.1 months and 63.5 ± 44.9 months in TAU and exercise groups.	60-minute sessions twice per week for 8 weeks including cycling, aerobic exercises and Tai-chi.	Treatment as usual.
McCartney et al., (2021)	Two-armed RCT. Clients receiving inpatient treatment for SUD were randomised to a week of exercise or stretching.	Exercise = 19; Control = 12	ICD cannabis dependant clients aged 18 + .(Exercise: 63% male, mean age 31, BMI 24.5 ± 5.1; con: 50% male, mean age 36, BMI 24.2 ± 3.3).	Acute: one week hospitalisation period.	Daily exercise for 4 days for 25 mins at 60% VO2max.	Daily stretching activities for 4 days for 25 mins.
Shimada et al., (2019)	Two-armed RCT. Inpatients with schizophrenia were randomised to 12 weeks of AE or TAU.	Exercise = 15; Control = 16	DSM-5 diagnosis of schizophrenia and aged 20–65 years.	No information given.	Aerobic exercise twice weekly for 12 weeks at 60–80% of aerobic capacity using a bike, treadmill and physical exercises.	Treatment as usual.
Shimada et al., (2020) (secondary analysis of Shimada et al., 2019)	Two-armed RCT. Inpatients with schizophrenia were randomised to 12 weeks of AE or TAU.	Exercise = 20; Control = 20	DSM-5 diagnosis of schizophrenia and aged 20–65 years. (Exercise: 43% male, body weight 61.35 ± 13.59 kg, age 50.14 ± 7.73; Con: 45% male, body weight 65.83 ± 8.79, age 49.75 ± 7.0).	No information given.	Aerobic exercise twice weekly for 12 weeks at 60–80% of aerobic capacity using a bike, treadmill and physical exercises.	Treatment as usual.
Imboden et al., (2020)	Two-armed RCT. Inpatients with MDD were randomised to 6 weeks of aerobic exercise (AE) or coordination/ stretching.	AE = 22; Active Control = 20	Inpatients aged>18 and<61 years with an ICD-10 diagnosis of depression who did not participate in regular vigorous-intensity exercise. (AE: 54.5% male, age 41.3 ± 9.2 years, BMI 25.9 ± 5.4 kg/m2; Con: 50% male, age 38.3 ± 13.4 years, BMI 23.9 ± 4.8).	Acute: The intervention period was determined by the typical length of patient hospitalization for severe depression.	6 weeks of supervised AE on indoor bicycles three times per week at 60–75% of HRmax, each session totalled 45 min.	6 weeks of basic coordination and stretching techniques for all major muscle groups using a medium-strength Theraband, a gymnastics ball, and juggling balls, three times per week.
Imboden et al., (2021) (secondary analysis of Imboden et al., 2020)	Two-armed RCT. Inpatients with MDD were randomised to 6 weeks of aerobic exercise (AE) or coordination/ stretching.	AE = 22; Active Control = 20	Inpatients aged>18 and<61 years with an ICD-10 diagnosis of depression who did not participate in regular vigorous-intensity exercise. (AE: 54.5% male, age 41.3 ± 9.2 years, BMI 25.9 ± 5.4 kg/m2; Con: 50% male, age 38.3 ± 13.4 years, BMI 23.9 ± 4.8).	Acute: The intervention period was determined by the typical length of patient hospitalization for severe depression.	6 weeks of supervised AE on indoor bicycles three times per week at 60–75% of HRmax, each session totalled 45 min.	6 weeks of basic coordination and stretching techniques for all major muscle groups using a medium-strength Theraband, a gymnastics ball, and juggling balls, 3 times per week.

(*Continued*)

Table 1. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/long stay)	Intervention	Control intervention
Gerber et al., (2020) (secondary analysis of Imboden et al., 2020)	Two-armed RCT. Inpatients with MDD were randomised to 6 weeks of AE or coordination/ stretching.	AE = 20; Active Control = 15	Inpatients aged>18 and<61 years with an ICD-10 diagnosis of depression who did not participate in regular vigorous-intensity exercise. (AE: male 57%, age 39.4 ± 9.7 years, BMI 24.7 ± 5.4 kg/m2; Con: 36% male, age 36.4 ± 14.8 years, BMI 22.6 ± 3.8).	Acute: The intervention period was determined by the typical length of patient hospitalization for severe depression.	6 weeks of supervised AE on indoor bicycles three times per week at 60–75% of HRmax, each session totalled 45 min.	6 weeks of basic coordination and stretching techniques for all major muscle groups using a medium-strength Theraband, a gymnastics ball, and juggling balls, 3 times per week.
Haussleiter et al., (2020)	Two-armed RCT. Inpatients with MDD were randomised to 6 weeks of guided exercise therapy (GET) or self-organized activity (SOA).	GET = 36; SOA = 40	Inpatients with a major depressive episode according to DSM-IV criteria. 66.7% were women. Mean age was 46.43 ± 11.60 and 43.94 ± 13.24 years in SOA and GET groups respectively, and mean BMI was 27.22 ± 5.58 and 26.05 ± 6.32.	A third of the initial inpatient cohort was discharged from the hospital within the study period, this may be suggestive of an acute ward setting.	3 GET sessions per week for 6-weeks, each session lasting 50 mins and comprised endurance training, body awareness, relaxation, group matches and activities.	The SOA group were encouraged to perform physical activity 3 times a week.
Buschert et al., (2019)	Two-armed RCT. Inpatients with depression were randomised to 3–4 weeks of physical exercise or occupational therapy (OT).	Exercise = 18; OT = 20	Inpatients diagnosed with unipolar depression without psychotic symptoms according to ICD-10 criteria. (Exercise: age 47.27 ± 6.84 years, 40% male; OT: age 47.47 ± 8.47 years, 33.3% male).	Depressed inpatients, 18.87 ± 13.04 and 15.80 ± 9.9 days since admission in exercise and OT groups respectively.	Endurance training (walking/running) of 30 min each, 2–3 times/week for 3–4 weeks. Exercise intensity was adapted to the individual patient's fitness level and 85% HRmax was the upper limit.	The OT group received 2–3 times per week (30 min) additional sessions of occupational or art therapy to improve sensory integration or gross motor skills (e.g., clay, wood).
Roy et al., (2018)	Two-armed RCT. Female inpatients with depression were randomised to 10 sessions of aerobic exercise or treatment as usual.	Exercise = 20; Control = 20	Women inpatients aged 18–60 years (Exercise 31.1 ± 8.88 years; Control 32.9 ± 8.53 years) with an ICD-10 diagnosis of depressive episode (70% subjects had severe depression and 25% moderate depression).	A tertiary care centre, Bengaluru, India.	The exercise group received a Video Assisted Structured Aerobic Exercise Programme every evening for 10 days. It consisted of 20 min of moderate/low intensity aerobic exercises. The video was played on laptop.	Treatment as usual.
Hanssen et al., 2018a	Two-armed RCT. Inpatients with a diagnosis of MDD were randomly assigned to 4 weeks of HIIT or MICT.	HIIT = 19; MICT = 15	Inpatients with a clinical diagnosis of MDD, without psychotic features, according to ICD-10 criteria. Mean baseline BDI-II score was 28.8 in the HIIT group and 33.8 in the MICT group, 26% and 27% were male in each group, mean age was 38.1 ± 12.2 and 37.5 ± 10.1 years and mean BMI was 22.6 ± 3.3 and 24.9 ± 5.2 in each group.	Acute: Intervention duration was limited by the duration of stay at the clinic.	HIIT thrice weekly for 4 weeks on a bicycle ergometer. HIIT consisted of 25 repetitions of 30-sec high intensity intervals at 80% VO2max followed by 30 sec of rest.	MICT thrice weekly for 4 weeks on a bicycle ergometer. MICT consisted of cycling for 20 mins at 60% VO2max
Hanssen et al., 2018b	Two-armed RCT. Inpatients with a diagnosis of MDD were randomly assigned to 4 weeks of HIIT or MICT.	HIIT = 12; MICT = 11	Inpatients with a clinical diagnosis of MDD, without psychotic features, according to ICD-10 criteria. Mean baseline BDI-II score was 27.9 in the HIIT group and 33.8 in the MICT group. Mean age was 38.7 ± 13.0 in the HIIT group, 34.6 ± 9.6 in the MICT group, 16.7% and 18.2% were male in each group and mean BMI was 22.8 ± 3.9 and 24.3 ± 5.5.	Acute: Intervention duration was limited by the duration of stay at the clinic.	HIIT thrice weekly for 4 weeks on a bicycle ergometer. HIIT consisted of 25 repetitions of 30-sec high intensity intervals at 80% VO2max followed by 30 sec of rest.	MICT thrice weekly for 4 weeks on a bicycle ergometer. MICT consisted of cycling for 20 mins at 60% VO2max.

(*Continued*)

Table 1. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/long stay)	Intervention	Control intervention
Minghetti et al., (2018)	Two-armed RCT. Inpatients with a diagnosis of MDD were randomly assigned to 4 weeks of SIT or MICT.	SIT = 29; MICT = 30	Inpatients aged 18–55 with a clinical diagnosis of MDD, without psychotic features, according to the ICD-10 criteria. Mean baseline BDI-II score was 31 ± 10 in the SIT group and 35 ± 10 in the MICT group. Mean age of 35 ± 12 years in SIT group and 37 ± 10 years in the MICT group, 27.6% and 20% were male in each group and mean BMI was 22.8 ± 3.4 and 24.9 ± 4.4.	Acute: Intervention duration was limited by the duration of stay at the clinic.	SIT performed thrice weekly for 4 weeks on a bicycle ergometer. SIT consisted of 25 repetitions of 30-sec high intensity intervals at 80% VO2max followed by 30 sec of complete rest.	MICT 3 times weekly for 4 weeks on a bicycle ergometer. MICT consisted of cycling for 20 mins at 60% VO2max.
Gerber et al., (2018)	Two-armed RCT. Inpatients with a diagnosis of MDD were randomly assigned to 4 weeks of SIT or MICT.	SIT = 25; MICT = 25 (with complete data)	Inpatients, with BMI≥30, and a clinical diagnosis of a major depressive episode according to ICD-10 criteria were recruited. (SIT: age 36.4 ± 12.4, BMI 23.0 ± 3.5, 24% male; MICT: age 36.5 ± 10.4, BMI 24.6 ± 4.4, 20% male).	Acute: Intervention duration was limited by the duration of stay at the clinic.	SIT performed thrice weekly for 4 weeks on a bicycle ergometer. SIT consisted of 25 repetitions of 30-sec high intensity intervals at 80% VO2max followed by 30 sec of complete rest.	MICT 3 times weekly for 4 weeks on a bicycle ergometer. MICT consisted of cycling for 20 mins at 60% VO2max.
Schneider et al., (2016)	Two-armed RCT. Inpatients with AUD were randomised to 3 weeks of Association Splitting (AS) or Exercise Therapy (ET).	AS = 30; ET = 34	Inpatient aged 18–75 years with a diagnosis of alcohol dependence, verified by the MINI International Neuropsychiatric Interview, Alcohol Use Disorders Test and Munich Alcoholism Test. (AS: age 45.53 ± 12.31 years, 66.7% male; ET: age 46.56 ± 13.37 years, 69% male).	Acute: Newly admitted patients receiving a standard 3-week inpatient treatment programme offered through the Addictive Disorders Clinic.	3× 90-min group sessions over 3 weeks. The Association Splitting model was explained and applied to craving and the patient's own alcohol-related cognitions. Keywords are identified that can cause or maintain craving (e.g., vodka, wine, relaxation, etc.). The group were asked to find neutral or positive associations to identified keywords.	ET was utilized as an active control group. ET was comprised of three 60-min sessions weekly over 3 weeks with low to high intensity physical activity (e.g., walking).
Legrand et al., 2016	Three-armed RCT. Inpatients with MDD were randomised to AE, placebo (stretching) (ST), or no intervention for 10 days.	AE = 14; ST = 11; Control = 10	Inpatients with a DSM-IV diagnosis of MDD were recruited. (AE: age 45.3 ± 10.6 years, 36.7% male; ST: age 41.8 ± 13.2 years, 27.3% male; Con: age 49.1 ± 16.5 years, 30% male).	Inpatients with MDD. The short length of the exercise programme may be suggestive of an acute stay.	AE consisted of 30 min of daily brisk walking or jogging outdoors for 10 consecutive days at 65–75% of HRmax.	ST consisted of a daily 30 min exercise programme of stretching exercises for 10 consecutive day. Muscle groups were stretched for 60s, with resting intervals between stretching. The control group received no intervention other than prescribed medication.
Loh et al., 2016	Two-armed RCT. Inpatients with schizophrenia were randomised to 3 months of walking or treatment as usual.	Exercise = 52; Control = 52	Inpatients aged 18–65 years with a DSM-IV diagnosis of schizophrenia (exercise: 67% male, age 46.0 ± 14, BMI 24.9 ± 6.9; con: 75% male, age 53.0 ± 11, BMI 23.93 ± 7.1).	Long stay: mean hospital stay was 9.43 ± 13.18 years.	20–30 minutes of walking at increasing intensity thrice weekly for three months.	Treatment as usual.
Salehi et al., (2016)	Three-armed RCT. Inpatients with MDD were randomised to 4 weeks of ECT, ECT & exercise or exercise.	ECT = 20; ECT + Exercise = 20; Exercise = 20	Inpatients aged 25–40 years with a DSM diagnosis of unipolar depressive disorder. (ECT: 65% male, age 29.70 ± 6.28; ECT + exercise: 65% male, age 30.25 ± 6.21; exercise: 80% male, age 29.05 ± 5.11).	No information given.	The intervention group received ECT + exercise. ECT was carried out at a Pulse width of 1.0 ms; seizure threshold was initially 50.4 mC(mC) and stepwise increased. ECT was administered three times a week for four consecutive weeks for both ECT and ECT + exercise groups.	Control groups received either ECT or exercise alone. Treadmill exercise at 60–75% VO2max for 40–45 mins thrice weekly for 4 weeks.

(*Continued*)

Table 1. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/long stay)	Intervention	Control intervention
Schuch et al., (2015)	Two-armed RCT. Inpatients with MDD were randomised to physical exercise or a non-active control until discharge.	Exercise = 25; Control = 25	Inpatients aged 18–60 years (Exercise: age 38.84 ± 11.5 years; Con: age: 41.76 ± 10.4 years) with a DSM-IV diagnosis of MDD who were not involved in other physical activity programmes (Exercise: 28% male, BMI 24.85 ± 5.2; Con: 24% male, BMI 25.46 ± 3.5).	Acute: The intervention period was from baseline to hospital discharge. The mean hospitalization length was 23.36 ± 9.0 days for the exercise group and 21.32 ± 8.2 days for the usual care group.	16 kcal/kg/week at 3 individual sessions/week in association with conventional treatment. Patients had the choice of a stationary bicycle, a treadmill or an elliptic and the intensity was chosen to complete the total calories estimated.	They were instructed to not perform exercise or stretching.
Schuch et al., (2014) (secondary analysis of Schuch et al., 2015)	Two-armed RCT. Inpatients with MDD were randomised to physical exercise or a non-active control until discharge.	Exercise = 15; Control = 11	Inpatients aged 18–60 years (Exercise: age 42.8 ± 12.4 years; Con: age: 42.5 ± 13.5 years) with a DSM-IV diagnosis of MDD who were not involved in other physical activity programmes (Exercise: 27% male, BMI 24.78 ± 4.4 kg/m2; Con: 28% male, BMI 25.52 ± 3.0).	Acute: The intervention period was from baseline to hospital discharge. The mean hospitalization length was of 21.63 ± 4.5 and 23.82 ± 5.7 days for exercise and control groups, respectively.	16 kcal/kg/week at 3 individual sessions/week in association with conventional treatment. Patients had the choice of a stationary bicycle, a treadmill or an elliptic) and the intensity was chosen to complete the total calories estimated.	They were instructed to not perform exercise or stretching.
Kerling et al., (2015)	Two-armed RCT. Inpatients with MDD were randomised to 6 weeks of exercise training or TAU.	Exercise = 22; TAU = 20	Inpatients aged 18 or over with a DSM-IV diagnosis of MDD receiving treatment in specialized psychotherapy wards and CBT. (Exercise: 55% male, age 44.2 ± 8.5 years, BMI 26.8 ± 5.1; TAU: 70% male, age 40.9 ± 11.9 years, BMI 26.8 ± 4.8).	Inpatients with moderate to severe depression. The short length of the exercise programme may be suggestive of an acute stay.	3× 45 min exercise sessions per week for 6 weeks at moderate intensity. Sessions were conducted on a bike, cross-trainer, stepper, arm ergometry, treadmill, recumbent or a rowing ergometry.	The daily activity programme of the ward (walking, ball games and stretching exercises for 20 min).
Kerling et al., (2017) (secondary analysis of Kerling et al., (2015))	Two-armed RCT. Inpatients with MDD were randomised to 6 weeks of exercise training or TAU.	Exercise = 22; TAU = 20	Inpatients aged 18 or over with a DSM-IV diagnosis of MDD receiving treatment in specialized psychotherapy wards and CBT. (Exercise: 55% male, age 44.2 ± 8.5 years, BMI 26.8 ± 5.1; TAU: 70% male, age 40.9 ± 11.9 years, BMI 26.8 ± 4.8).	Inpatients with MDD. The short length of the exercise programme may be suggestive of an acute stay.	3× 45 min exercise sessions per week for 6 weeks at moderate intensity. Sessions were conducted on a bike, cross-trainer, stepper, arm ergometry, treadmill, recumbent or a rowing ergometry.	The daily activity programme of the ward (walking, ball games and stretching exercises for 20 min).
Kerling et al., (2018) (secondary analysis of Kerling et al., 2015)	Two-armed RCT. Inpatients with MDD were randomised to 6 weeks of exercise training or TAU.	Exercise = 20; TAU = 10	Inpatients aged 18–60 with a DSM-IV diagnosis of MDD receiving treatment in specialized psychotherapy wards and CBT. (Exercise: 60% male, age 44 ± 8 years, BMI 26.3 ± 5.0; TAU: 60% male, age 38 ± 9 years, BMI 26.5 ± 3.3).	Inpatients with MDD. The short length of the exercise programme may be suggestive of an acute stay.	3× 45 min exercise sessions per week for 6 weeks at moderate intensity. Sessions were conducted on a bike, cross-trainer, stepper, arm ergometry, treadmill, recumbent or a rowing ergometry.	The daily activity programme of the ward (walking, ball games and stretching exercises for 20 min).
Kahl et al., (2016) (secondary analysis of Kerling et al., (2015))	Two-armed RCT. Inpatients with MDD were randomised to 6 weeks of exercise training or TAU.	Exercise = 20; TAU = 10	Inpatients aged 18–60 with a DSM-IV diagnosis of MDD receiving treatment in specialized psychotherapy wards and CBT. (Exercise: 60% male, age 43.8 ± 8.1 years, BMI 26.3 ± 5.0; TAU: 60% male, age 38.4 ± 9.3 years, BMI 26.5 ± 3.3).	Inpatients with MDD. The short length of the exercise programme may be suggestive of an acute stay.	3× 45 min exercise sessions per week for 6 weeks at moderate intensity. Sessions were conducted on a bike, cross-trainer, stepper, arm ergometry, treadmill, recumbent or a rowing ergometry.	The daily activity programme of the ward (walking, ball games and stretching exercises for 20 min).

(*Continued*)

Table 1. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/long stay)	Intervention	Control intervention
Rosenbaum et al., (2015)	Two-armed RCT. Inpatients with PTSD were randomised to 12 weeks of exercise or usual care.	Exercise = 39; Usual Care = 42	Inpatients aged over 18 years receiving a PTSD programme at a private hospital with a DSM-IV diagnosis of primary PTSD. (Exercise: age 47.1 ± 11.3 years, 92% male, BMI 29.0 ± 4.8; Con: age 52.0 ± 12.7 years, 76% male, BMI 31.7 ± 4.7).	Acute: Average length of stay for the inpatient programme was 3 weeks.	The 12-week intervention consisted of a weekly supervised exercise session, two unsupervised home-sessions, and a walking programme. The exercise protocol consisted of three sets of 10 repetitions of up to six exercises, with a rest period of between 30–60 s between sets. Moreover, participants were given a pedometer and encouraged to aim for a daily target of 10 000 steps.	Both groups received usual care involving a combination of psychotherapy, pharmaceutical interventions, and group therapy facilitated by psychologists.
Heissel et al., (2015)	Two-armed RCT. Older inpatients with MDD were allocated to 4 weeks of physical exercise (aerobic, strength, and coordination exercises) or an active control (relaxation exercises).	Exercise = 6; Control = 6	Inpatients with a DSM-IV diagnosis of MDD. The mean age was 68.7 ± 3.1 and 64.8 ± 9.1 years in exercise and control groups respectively and 33.3% and 50% were male in each group.	Acute: an average hospital stay of 5–6 weeks.	4 week supervised programme, twice a week for 60 minutes. The group included walking exercises for 20 to 25 min (e.g., walking with a ball or through an obstacle course or through the park), strength and strain exercise (10 min) and balance tasks (e.g., single leg stance or bipedal leg stance).	4 week supervised programme, twice a week for 60 min. Participants practiced several relaxation tasks (e.g., progressive muscle relaxation) and stretching of the main muscle groups of the legs and arms, shoulder and neck.
Shachar-Malach et al., 2015	Two-armed RCT. Inpatients with MDD were randomised to 3 weeks of AE or stretching.	AE = 6; Stretching = 6	Inpatients (aged 18–80) with a DSM-IV diagnosis of MDD. The mean age was 33.17 ± 13.41 years and 53.50 ± 18.78 years in AE and stretching groups and 33.4% and 16.7% were male in each group.	Acute: mean length of hospitalization was 38.83 +25.36 days and 61.16 + 52.4 days in AE and stretching groups respectively.	4 individual AE sessions per week for 3 weeks consisting of 30 min walking on a treadmill with a moderate intensity corresponding to 60–80% of HRmax.	4 individual stretching sessions per week for 3 weeks. Stretching exercises were performed a very low intensity.
Kim et al., (2014)	Two-armed RCT. Male inpatients with schizophrenia, depression and Bipolar disorder were randomised to 12 weeks of exercise or standard care.	Exercise = 23; Control = 19	Mae inpatients with a DSM-IV psychiatric diagnosis. In the exercise group 69.6% had schizophrenia, 21.7% depression and 8.7% Bipolar disorder. In the control group, 63.2% had schizophrenia, 21.1% depression and 15.8% Bipolar disorder. (Exercise: age: 46.4 ± 2.1 years, BMI 25.2 ± 1.1; Control: age 50.8 ± 2.6 years, BMI 24.3 ± 1.1).	Mentally ill patients admitted to closed hospital wards for extended stays.	60 min aerobic sessions, three days a week for 12 wk. Each session included 2 × 10 min of stretching and 40 min of aerobic activity using a combination of walking, running and other aerobic movements to achieve 50% HRR between weeks 1–4, 60% HRR between weeks 5–8, and 70% HRR between weeks 9–12.	Ordinary daily activities.
Kim et al., 2014b	Two-armed RCT. Inpatients with schizophrenia took part in 12 weeks of exercise or control.	Exercise = 24; Control = 12	Patients with paranoid schizophrenia (DSM-IV diagnosis) who had been chronically hospitalized more than 3 years in psychiatry ward. (Exercise: age 48.7 ± 9.9 years, BMI 25 ± 4.3; Control: age 50.7 ± 12.0 years, BMI 26.5 ± 4.7).	Long stay: patients had been chronically hospitalized for more than 3 years.	60 min of combined exercise 3 days per week for 12 weeks consisting of strength training and walking for 25 min at 50–70% of HRR. The strength training routine consisted of: chest press, seated row, squat, shoulder press, biceps curl, triceps extension, calf raise, and reverse crunch.	Routine daily activities and performed stretching (static and dynamic), dancing and recreation activity once a week for 1 h.

(*Continued*)

Table 1. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/long stay)	Intervention	Control intervention
Ho et al., 2014	Two-armed RCT. Inpatients with mild-to-moderate depression took part in 3 weeks of AE or control.	Exercise = 26; Control = 26	Inpatients, aged 18–64 years, with mild to moderate depression who did not partake in regular exercise. (Exercise: age 43.62 ± 13.3 years, 35% males, BMI 22.33 ± 3.31; Control: age 48.81 ± 11.30 years, 31% males, 23.22 ± 4.61).	Eleven subjects dropped out of the 3-week intervention due to discharge, this may be suggestive of an acute ward setting.	5 supervised interval-training style exercise sessions per week for 3 weeks, including both upper extremity and lower extremity aerobic exercise training. Sessions included 30-min interval training, consisting of 3 bouts of 5-min workout at 40–59% HRR.	Subjects in control group maintained their physical activity level as usual. A 10-min stretching exercise on large muscle groups was given.
Manjunath et al., (2013)	Two-armed RCT. Inpatients with non-affective psychosis were randomised to 6 weeks of yoga or physical exercise.	Yoga = 44; Exercise = 44	Newly admitted (within past 1 week) patients with non-affective psychosis. All except five had a diagnosis of schizophrenia. (Yoga: age 31.7 ± 8.8 years, 59% male, exercise: age 31.1 ± 7.8 years, male 52% male).	Acute: Newly admitted patients participating in 2 weeks of in-patient treatment.	Each yoga session lasted 1 h. After 2 weeks, patients practiced the same for the next 4 weeks.	Daily one-hour sessions over 2 weeks in the wards including brisk walking, jogging and exercises in standing and sitting postures. After 2 weeks, patients practiced the same for the next 4 weeks (after discharge).
Knubben et al., (2007)	Two-armed RCT. Inpatients with MDD were randomly assigned to 10 days of walking or a control.	Walking = 20; Control = 18	Inpatients aged 20–70 years receiving treatment for a major depressive episode (DSM-IV). (Exercise: age 49 ± 13 years, 45% male; Control: age 50 ± 13 years, 44% male).	Acute: Mean length of hospitalization was 47 ± 2 and 58 ± 38 days in walking and control groups.	Daily interval training walking on a treadmill for 10 days for 30-min. Patients walked five times for 3 min at 80% HRmax interspersed with 3 min rests at half speed (Borg rating 13–14).	A daily 30-min programme for 10 days consisting of light stretching exercises for the calves, thighs, back, shoulders, pectoral muscles, and relaxation exercises. Each muscle group was stretched for 20s, with a resting interval of 40s between stretching series.
Knapen et al., (2003)	Two-armed RCT. Non-psychotic inpatients were randomised to a 16 week personalized psychomotor fitness programme (PF) or a general programme of psychomotor therapy (GPMT).	PF = 102 (57 completed follow-up assessments); GPMT = 97 (51 completed follow-up)	Inpatients with a long psychiatric history, with severe depressive and/or anxious symptoms, and/or personality disorders (DSM-IV diagnosis). Patients were excluded if they had psychosis. The groups did not differ in age (PF: 35.60 ± 10.71 years; GPMT: 32.02 ± 10.32 years).	Patients were newly admitted to specialised treatment units. 33 patients in the exercise group were discharged at week 8 and 19 prior to the 16 week end of intervention.	Endurance, strength and flexibility training, thrice weekly for 16 weeks at 40–60% HRR. The exercise units were the treadmill, the multistation resistance machine, the rowing machine, the bicycle ergometer and the step machine. Each session lasted 45 min, including stretching.	Physical exercises, twice a week, and progressive relaxation training once a week for 16 weeks, each lasted for 45 min. Physical activities at low to moderate intensity, included moving to music, badminton, gymnastics, body awareness techniques and sports and games (e.g., volleyball, basketball, indoor hockey, soft-tennis).
Knapen et al., (2005) (secondary analysis of Knapen et al., (2003))	Two-armed RCT. Non-psychotic inpatients were randomised to a 16 weeks personalized psychomotor fitness programme (PF) or a general programme of psychomotor therapy (GPMT).	PF = 102 (57 completed follow-up assessments); GPMT = 97 (51 completed follow-up)	Inpatients with a long psychiatric history (mean 7.78 ± 9.21 years) with severe depressive and/or anxious symptoms, and/or personality disorders (DSM-IV diagnosis). Patients were excluded if they had psychosis. The mean age of the PF group (35.54 ± 10.76) was significantly higher than that of the GPMT group (32.44 ± 10.75) (p = 0.04).	Patients were newly admitted to specialised treatment units. 39 patients were discharged at week 8, and 52 prior to the 16-week end of intervention.	Endurance, strength and flexibility training, thrice weekly for 16 weeks at 40–60% HRR. The exercise units were the treadmill, the multi-station resistance machine, the rowing machine, the bicycle ergometer and the step machine. Each session lasted 45 min, including stretching.	Physical exercises, twice weekly, and progressive relaxation training once a week for 16 weeks, each lasted for 45 min. Physical activities at low to moderate intensity, included moving to music, badminton, gymnastics, body awareness techniques and sports and games (e.g., volleyball, basketball, indoor hockey, soft-tennis).

(*Continued*)

Table 1. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/long stay)	Intervention	Control intervention
Palmer et al., (1995)	Three-armed RCT. Inpatients with SUD or AUD were randomly assigned to either 4 weeks of aerobics, bodybuilding, or circuit training.	aerobics = 15; bodybuilding = 15; Circuit training = 15	Inpatients, aged 18–55 years (mean 28 yrs), undergoing a 28- to 45-day rehabilitation programme at a state facility for problems with alcohol, cocaine, or other drugs. 76% were male.	Acute: Subjects were undergoing a 28-to 45-day rehabilitation programme.	Subjects exercised three times per week for 30–40 min. The aerobic step group stepped up and down on benches at 60% HRmax.	The Circuit training group made two circuits around nine exercise stations: bench press, leg curl, lat pull, leg press, abdominal crunch, incline press, stationary bicycle, biceps curl, and stepper machine. The bodybuilding group performed three sets of the following exercises: bench press, leg curl, lat pull, leg press, abdominal crunch, incline press, arm curl, seated row, triceps extension, and toe press.
Bosscher, 1993	Two-armed RCT. Inpatients with depression/mood disorder were randomised to 8 weeks of running therapy or TAU (mixed physical and relaxation exercises).	Exercise = 9; TAU = 9	Inpatients with depression/mood disorders who had been hospitalised for 1–4 months in a psychiatric ward. The mean age was 35.8 ± 11.1 years in the running group (5 men, 4 women) and 34.7 ± 12.5 in the TAU group (4 mean, 5 women). Patients who took antidepressants/neuroleptics were excluded.	At the start of the intervention, participants had been hospitalised for 1–4 months	Running thrice week for 8 weeks for 45 mins per session. Sessions included a 10-min warm-up with stretching, a 30 min running phase at 70–85% HRmax, and a 5-min cool-down.	The TAU group were offered sports (volleyball, soccer), gymnastics, trampolining twice weekly. Sessions lasted 50 min. The emphasis was relaxation and low-intensity physical activity. Patients were also offered weekly 45-min relaxation and breathing exercises.
Sexton et al., (1989)	Two-armed RCT. Inpatients with neurotic symptoms were randomly assigned to 8 weeks of walking or jogging.	Walkers = 25; Joggers = 28	Inpatients, aged 20–60 years, with an inpatient stay over 2 weeks, and a nonpsychotic DSM-I diagnosis. (Walkers: age 39.2 (19–60), 40% male, 8 anxiety diagnosis, 14 depressive diagnosis, 3 substance abuse; Joggers: age 36.4 (19–53), 54% male, 13 anxiety diagnosis, 11 depressive diagnosis, 8 substance abuse).	Acute: short-stay nonpsychotic inpatients with a stay over 2 weeks.	The groups exercised 30 min 3–4 times per week for 8-weeks.	Joggers were instructed to jog at a level commensurate with their physical capacity (approximately 70% HRmax). Walkers walked at a comfortable speed.
Martinsen et al., (1989)	Two-armed RCT. Inpatients with depression were randomly assigned to 8 weeks of aerobic, or non-aerobic, exercise.	Aerobic = 51; Non-aerobic = 47	Inpatients meeting the DSM-III criteria for major depression, dysthymic disorder, or depressive disorder not otherwise specified. (Aerobic: age 40.9 ± 9.1 years, 35% male; Non-aerobic: age 41.2 ± 10.9 years, 36% male).	No information given.	An hour of training, thrice weekly for 8 weeks. The aerobic group performed brisk walks/jogging at 70% of max aerobic capacity.	The nonaerobic group trained muscular strength, flexibility, and relaxation at low intensity.
Martinsen et al., 1989b	Two-armed RCT. Inpatients with anxiety disorders were randomly assigned to 8 weeks of aerobic, or non-aerobic, exercise.	Aerobic = 36; Non-aerobic = 43	Inpatients who met DSM-III criteria for one of the anxiety disorders: panic disorder, agoraphobia without panic attacks, social phobia or generalized anxiety disorder. (Aerobic: age 39.1 ± 8.1 years, 31% male; Non-aerobic: age 38.8 ± 8.5 years, 40% male).	No information given.	1 hour of training, thrice weekly for 8 weeks. The aerobic group performed brisk walks/jogging at 70% of max aerobic capacity.	The nonaerobic group trained muscular strength, flexibility, and relaxation at low intensity.

(*Continued*)

Table 1. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/long stay)	Intervention	Control intervention
Gary & Guthrie, 1972	Two-armed RCT. Male inpatients with alcohol abuse were randomised to 4 weeks of jogging or a non-active control.	Jogging = 10; Non-active control = 10	Male alcoholic patients on an alcoholic treatment ward with an average of 18 years of problem drinking. Mean age was 38.9 years (range 25–55 years) and 45.1 years (range 39–56 years) in exercise and control groups respectively.	No information given.	Jogging a mile course 5 days a week for 4 weeks or until a total of 20 miles had been reached.	The control group maintained normal activities including group psychotherapy, ward clean-up and recreation programmes.

Table 2. Basic characteristics of included non-RCTs including study design, population characteristics and details of exercise and control interventions.

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/ long stay)	Intervention	Control intervention
Durmus et al., 2020	Two-armed controlled study. Male inpatients with opiod use disorder (OPuD) completed 5 sessions of HIIT or control.	HIIT = 11; Control = 11	Male inpatients aged 18–45 years, diagnosed on the DSM-5 criteria with OpUD. Mean age was 27.0 ± 7.3 and 27.1 ± 5.2 years in HIIT and control groups respectively, and mean BMI was 21.8 ± 2.2 and 21.8 ± 2.1.	Acute: all patients were hospitalized for a 21 day stay.	5 bicycled based HIIT sessions over 21 days. Sessions included three 30 sec bursts of loading at high intensity interspersed with 4 min of resting on the 30W pedal.	No details of control activities given.
Kern et al., (2020)	A pilot pre-post-test intervention. Female inpatients with Anorexia Nervosa (AN) took part in an eight-session adapted physical activity programme (APA).	APA = 29	Female inpatients, aged 13–20 years old (mean 16.4 ± 1.3 years old) with AN. Mean BMI was 16.8 ± 2. Fifty six percent reported rarely participating in physical activity.	Several participants did not complete the programme due to hospital discharge, this may be suggestive of an acute ward setting.	8× 90 min exercise sessions were offered, once per week. Sessions included a 25 min muscular workout, 25 mins of shadow French boxing, a 10 min cool down, and a welcome and debriefing.	
Stanley et al., (2019)	Pre-post-test intervention. Mental health inpatients took part in the 12 week long "Fit for Life" programme.	Fit for Life = 72	Mental health inpatients, 39 females (mean age 43.1 ± 12.7 years) and 33 male (mean age 43.3 ± 12.6 years). Primary diagnostic categories were: males 51.5% psychotic disorder and 39.4% mood disorder; females 48.7% mood disorders, 18% personality disorders, and 18% neurotic disorders.	No information given.	A physiotherapist conducted a fitness assessment and directed participants towards the general gym programme or individualised exercise programme. Participants attended the gym at least twice a week.	
Mazyarkin et al., (2019)	Controlled pilot study. Mental health inpatients exercised for up to 6 months, matched patients served as a control.	Exercise = 23; Control = 10	Adults aged 18–60 years receiving inpatient treatment, all of the patients except one patient with bipolar disorder, were diagnosed with schizophrenia or schizoaffective disorder according to the DSM-V. (Control: age 39.5 ± 1.7 years, 100% male, BMI 29.2 ± 1; Exercise up to 3 months; age 35.5 ± 2.7 years, 91% male, BMI 26.9 ± 2; Exercise over 3 months: age 32.5 ± 2.4 years, 100% male, BMI 26.9 ± 1).	Patients recruited from open and closed wards.	Subjects trained for up to 6 months, twice a week for 45 min. The training programme included aerobic (long walks, jogging, and short running intervals) and resistance training (TRX-straps-workout, push-ups, pull-ups, and sit-ups). The level of difficulty increased from week to week.	Patients in the control group did not take part in the exercise programme.
Wunram et al., (2018)	Three-armed semi randomised trial (control group were not randomised). Adolescent inpatients with depression completed 6 weeks of passive muscular training on a whole-body vibration (WBV), exercise training or control.	Exercise = 20; WBV = 21; Control = 23	Inpatients or day-clinic patients aged 12–18 years (89.1% of the total sample were inpatients) who met DSM-IV/-5 and ICD-10 criteria for MDD. (Exercise: age 16.1 ± 1.2 years, 40% male, BMI 26 ± 7.6; WBV: age 15.9 ± 1.2 years, 28.6% male, BMI 24.7 ± 5.9; Con: age 15.7 ± 1.1 years, 17.4% male, BMI 23.3 ± 5.0).	Acute: Mean length of hospital stay was 80 ± 36 days, 60 ± 20 days and 64 ± 39 days in exercise, WBV and control groups respectively.	The WBV group completed 18 exercise sessions in 6 weeks on the Galileo training device (frequency of 20 Hz and an amplitude of 2 mm). The vibration plate stimulates a movement pattern similar to human gait.	The exercise group received 18 exercise sessions in 6 weeks. Ergometer training on stationary cycles consisting of a 30-min interval training regime. Participants in the control condition received treatment as usual.

(*Continued*)

Table 2. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/ long stay)	Intervention	Control intervention
Wunram et al., (2021) (secondary analysis of Wunram et al., 2018)	Three-armed semi randomised trial (control group were not randomised). Adolescent inpatients with depression completed 6 weeks of passive muscular training, exercise training or control.	Exercise = 20; WBV = 21; Control = 23	Inpatients or day-clinic patients aged 12–18 years (89.1% of the total sample were inpatients) who met DSM-IV/-5 and ICD-10 criteria for MDD. (Exercise: age 16.1 ± 1.2 years, 40% male, BMI 26 ± 7.6; WBV: age 15.9 ± 1.2 years, 28.6% male, BMI 24.7 ± 5.9; Con: age 15.7 ± 1.1 years, 17.4% male, BMI 23.3 ± 5.0).	Acute: Mean length of hospital stay was 80 ± 36 days, 60 ± 20 days and 64 ± 39 days in exercise, WBV and control groups respectively.	The WBV group completed 18 exercise sessions in 6 weeks on the Galileo training device	The exercise group received 18 exercise sessions in 6 weeks. Ergometer training on stationary cycles consisting of a 30-min interval training regime. Participants in the control condition received treatment as usual.
Song et al., (2018)	Two-armed controlled trial. Male psychiatric inpatients took part in 12 weeks of exercise or control.	Exercise = 31; Control = 21	Male inpatients with mental illness who did not participate in regular exercise. (Exercise: mean age 48.8 ± 9.5 years, BMI 25.8 ± 4.4; Control: age 49.9 ± 11.2, BMI 24.1 ± 4.1).	A closed ward lack because of long term hospitalization.	12 weeks of combined exercise thrice weekly. Elastic band exercises and a running programme were used for resistance exercise and aerobic exercise, respectively. Resistance training intensity increased from 10 RM to 15 RM from weeks 2 to 3. Aerobic exercise was conducted at 60% VO2max.	No details of control group activities given.
Woodward et al., (2018)	Longitudinal pre-post investigation. Inpatients with treatment resistant schizophrenia were randomised to an aerobic or weight bearing exercise programme for 12 weeks and results from both treatment arms were analysed together.	Aerobic = 9; Weights = 8	Treatment-resistant schizophrenia inpatients aged 19–50 years with no cardiovascular or mobility issues (mean age 30.1 ± 6.9, 59% male, mean BMI 28.5 ± 2.2). All participants had been receiving treatment for psychosis for a minimum of 36 months at baseline.	Most people stay in the Psychosis Programme for an average of four to six months.	Three sessions weekly for 12 weeks consisting of 30 min of moderate-intensity exercise at 40–59% of HRR using different modalities (cycle ergometry, treadmill, elliptical training and weight training).	30 min of weight training based on their individual muscular capacity thrice weekly.
Kerling et al., (2016)	Pre-post-test intervention with control group. Inpatients with MDD participated in 6 weeks of exercise training, they were matched to 25 healthy controls.	Exercise = 25; Control = 25	Inpatients aged 18–65 with a DSM-IV diagnosis of MDD (age 45.4 ± 9.1 years, BMI 26.4 ± 5.3, 48% male). Patients were matched with a healthy control group for age, gender, and BMI (age 45.4 ± 10.6 years, BMI 26.4 ± 3.9, 48% male).	Inpatients with MDD. The short length of the exercise programme may be suggestive of an acute stay.	3× 45 min exercise sessions per week for 6 weeks at moderate intensity (Borg rating 13–14). Sessions were conducted on a bike, cross-trainer, stepper, arm ergometry, treadmill, recumbent or a rowing ergometry.	The control group did not take part in the exercise intervention.
Heggelund et al., (2011)	Non-randomized controlled trial. Inpatients with schizophrenia were allocated to either 8 weeks of HIIT or a computer game programme (CGP).	HIIT = 16; CGP = 9	Inpatients with an ICD-10 diagnosis of schizophrenia, schizotypal and delusional disorders and receiving stable antipsychotic medication for≥6 weeks. (HIIT: age 30.5 ± 8.7 years, 75% male, BMI 28.8 ± 4.7; CGP: age 38.9 ± 11.4 years, 57% male, BMI 27.6 ± 8.5).	Long stay: The majority of patients had several years of hospitalization (mean months of hospitalization: 35.2 ± 19.6 and 70.1 ± 62.9 in HIIT and CGP groups respectively).	8 weeks of supervised HIIT thrice weekly. HIIT consisted of 4 × 4-min intervals at 85–95% HRpeak interspersed with 3 min recovery periods at 70% HRpeak. Participants exercised on a treadmill with a minimum of 5% incline.	Supervised computer games training thrice weekly for 8 weeks. Training to the computer game Tetris using an Xbox Video Game Systems.
Ng et al., (2007)	Retrospective cohort study. Inpatients with Bipolar disorder who regularly attended an inpatient daily walking group were compared with those who did not attend.	Participant admissions = 24 (from 14 patients); Non-participant admissions = 74 (from 35 patients)	Patients admitted to a private psychiatric inpatient unit with a diagnosis (ICD-10) of bipolar disorder. (Participant admissions: 46% male, age 43.6 ± 15.0 years; Non-participant admissions: 15% male, age 43.9 ± 18.5 years).	Acute: Mean length of stay was 19.3 ± 14.0 and 18.6 ± 17.4 days in participant and non-participant admissions.	The walking group was scheduled on weekday mornings, and consisted of a nurse-led 40-min walk in the hospital vicinity. Regular participants were patients who reliably attended the group throughout their admissions.	Participants who did not attend the walking group during their admission.

(*Continued*)

Table 2. (Continued).

Study	Study design and sample included	Sample size	Sample characteristics	Type of inpatient setting (acute ward or rehabilitation/ long stay)	Intervention	Control intervention
Gaul-Alacova et al., 2005	Two-armed controlled trial. Inpatients with anxiety took part in a 6-week exercise programme or control group.	Exercise = 27; Control = 16	Anxiety patients (ICD-10 diagnosis F40.0-F41.2) hospitalized for 6 weeks of intensive psychotherapy treatment. (Exercise group: 19% male, mean age 33.50 ± 9.59 years; control: 56% male, mean age 32.76 ± 9.71 years).	Acute: patients were hospitalised for 6 weeks for an intensive psychotherapy programme.	The patients exercised 3 times a week for 30-mins on a stationary bicycle at individualised load after initial stretching.	The controls underwent an identical 6-week Integrative Therapeutic Psychotherapy Programme.
Martinsen et al., 1989c	Pre-post-test intervention. Mental health inpatients took part in an 8-week aerobic exercise regime.	Exercise = 92	Nonpsychotic inpatients who had not responded well to usual treatment (psychodynamic therapy) ($n = 41$) or patients with somatoform complaints who had previously not been motivated for psychiatric treatment ($n = 36$) or were on the waiting list ($n = 15$). Mean age was 39.8 ± 9.8 years.	Acute: all patients were hospitalized for 8 weeks.	1 hour of aerobic exercise five times a week for 8 weeks at 50–70% of max aerobic capacity. Walking and jogging were the commonest activities, but skiing, cycling, swimming, and aerobics were also used.	
Tsukue & Shohohi, 1981	A pre-post-test intervention. Male inpatients with alcohol abuse took part in a 10-month long basketball programme.	Exercise = 25	Male alcoholic residents with no abnormalities in physical tests (cardiac, liver, blood sugar). Age ranged from 26–69 years. Most participants were aged 41–45 years.	Long stay: all patients were hospitalized for at least 6 months at study commencement.	3 one-hour basketball sessions each week for 10 months, basketball sessions comprised running, jumping and throwing.	

Table 3. Mental health findings, additional outcomes, adherence and adverse events from included RCTs.

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Lan et al., (2022)	Adult Mental Health Scale (AMHS) scale	The two groups showed statistically significant differences in "somatic symptom" ($t = -4.53$, $p < 0.001$), "anxiety" ($t = -2.97$, $p = 0.004$), "depression" ($t = -2.18$, $p = 0.031$), "social distress" ($t = -2.08$, $p = 0.040$), and "positive optimism" ($t = 1.08$, $p = 0.028$) with significantly greater improvements in the health group.		Adverse events not reported.	160 people recruited and 40 dropped out during study implementations- 75% participation rate.	No support given to exercise post-discharge.	Exercise was associated with improvements in mental health compared to control.
Philippot et al., 2022	The Hospital Anxiety & Depression Scale (HADS); The Hamilton Depression Rating Scale (HAM-D); Child Depression Inventory (CDI); Beck's abbreviated Depression Inventory (BDI-13); Zung Self-Assessment Depression Scale (SDS); STAI; physical health	A significant interaction in favour of physical exercise in reducing the mean depression score (HADS-D) by 3.8 points, compared to a mean reduction score of 0.7 in the control group. Significant greater improvement in SDS score. No significant improvement in other depression, or anxiety measures.	These VO2max indices revealed a significant group x change interaction ($p = 0.024$) in favour of the physical exercise group.	Adverse events not reported.	Six participants were lost in each group (23%). The mean attendance rates of the exercise (76%) and control group (79%) did not differ. Participants completed a mean of 18.4 and 18.1 sessions (out of 20) in exercise and control groups.	No support given to exercise post-discharge; dropout reflected discharge.	Exercise may reduce depressive symptoms in hospitalised adolescents.
Kurebayashi et al., 2021	The Cognitrax Basic Version neurocognitive test; PANSS	Greater improvements in neurocognitive index ($p = 0.009$), psychomotor speed ($p = 0.037$), attention ($p = 0.011$), flexibility ($p = 0.007$), processing speed ($p = 0.008$) and executive function ($p = 0.016$). No change in PANNS scores.		No adverse events.	The dropout rates in the TAU and exercise groups were 17.6% and 20%, respectively.	No support given to exercise post-discharge; one patient dropped out due to discharge.	Mild-intensity exercise may improve neurocognition in schizophrenia inpatients.
McCartney et al., (2021)	Wrist actigraphy sleep data	Sleep onset latency increase in the control group ($p = .042$) whereas the Cycling group exhibited improvements in sleep duration($p = .008$) and sleep efficiency ($p = .023$) compared to Baseline, and increased sleep duration ($p = .005$), decreased average wake bout($p = .040$) compared to stretching.		Adverse events not reported.	No dropouts. Five sessions (<3.0%) were missed entirely.	No support given to exercise post-discharge.	Moderate-intensity aerobic exercise may attenuate the sleep disturbances associated with cannabis withdrawal.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Shimada et al., (2019)	Brief Assessment of Cognition in Schizophrenia (BACS); quality of Life Scale (QLS); PANSS; Scale for the Assessment of Negative Symptoms (SANS; Life Assessment Scale for the Mentally Ill (LASMI); GAF	Significant treatment-by-time interaction effects on working memory, verbal fluency, attention, executive function, and composite score in BACS; in- formant total in SCoRS; QLS score; positive, negative, and total in PANSS; affective flattening, avolition/apathy, anhedonia/asociality, attention, total in SANS; and interpersonal relations in LASMI.		No adverse events.	One patient dropped out of the TAU alone arm.	No support given to exercise post-discharge.	Aerobic exercise may improve cognitive functioning in schizophrenia.
Shimada et al., (2020) (secondary analysis of Shimada et al., 2019)	Brief Assessment of Cognition in Schizophrenia (BACS); quality of Life Scale (QLS); PANSS; Scale for the Assessment of Negative Symptoms (SANS; Life Assessment Scale for the Mentally Ill (LASMI); GAF	At 1-year follow-up, the TAU + AE demonstrated significant group by time interaction effects in several cognitive domains, intrinsic motivation, negative symptom, interpersonal relations, and functional outcome compared with the TAU alone.		No adverse events.	One patient (3.13%) dropped out of the TAU alone arm.	No support given to exercise post-discharge.	Improvements in cognition and other outcomes with aerobic exercise can be maintained over 1-year follow-up.
Imboden et al., (2020)	17-item Hamilton Depression Rating Scale (HDRS17); Beck Depression Inventory (BDI); Mental Toughness Questionnaire; Cognition; BP; HR; BMI; Cardiorespiratory fitness	There was a time effect for symptom-severity (HDRS17: $p < 0.001$, $\eta^2 = 0.70$; BDI: $p < 0.001$, $\eta^2 = 0.51$) but no significant time by group interaction (HDRS $p = 0.687$; BDI $p = 0.412$). Mental toughness improved in both groups ($p < 0.001$, $\eta^2 = 0.32$). Short-term effects on symptom severity and mental toughness remained stable across the 6-month follow-up.	Working memory showed a significant time by group interaction in favour of AE ($p = 0.043$, $\eta^2 = 0.10$). Cardiorespiratory fitness increased in both groups. No change in BP and resting HR.	No exercise related adverse events.	After randomisation 22.7% ($n = 5$) dropped out in the AE group and 19.0% ($n = 4$) in the active control group. 64.7% of participants completed all training sessions.	No support given to exercise post-discharge. Two patients with early discharge had to be excluded from the study.	AE was associated with depression alleviation, and add-on benefits on working memory compared to stretching.
Imboden et al., (2021) (secondary analysis of Imboden et al., (2021))	CAR, sBDNF and TNF-alpha; Pittsburgh Sleep Quality Index (PSQI); polysomnography (PSG)		Stress axis activity decreased in both groups from baseline to post-intervention. sBDNF showed a significant increase over time, whereas the number of awakenings significantly decreased. No significant time by group interactions were detected for any of the study variables.	No exercise related adverse events.	Attendance not reported.	No support given to exercise post-discharge. Two patients with early discharge had to be excluded from the study.	No evidence was found for differential effects between AE and an active control for neurobiological variables.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Gerber et al., (2020) (secondary analysis of Imboden et al., 2020)	Cortisol reactivity during the Trier social stress test (TSST)		Cortisol reactivity did not change from baseline to post-intervention, either in the intervention or control group.	No exercise related adverse events.	Fourteen AE participants and 12 control participants remained at post-intervention. 64.7% of participants completed all 18 training sessions.	No support given to exercise post-discharge. One patient with early discharge had to be excluded from the study.	AE was not associated with cortisol reactivity.
Haussleiter et al., (2020)	Depression (HAM-D); WHO Well-Being Index (WHO-5); The Personal and Social Performance (PSP) scale (functioning); physical fitness	GET was superior to SOA in reducing depression symptom severity ($p = 0.017$), specifically improving suicidality ($p = 0.028$) as well as time ($p = 0.003$) and severity of diurnal variation ($p = 0.027$).	A significant (time X group) interaction effect regarding the PSP total score indicated superiority of the GET over the SOA ($p = 0.002$). There was no time X group effect for WHO-5 total score, Timed Up-and-Go Test and the Sit-and-Reach Test.	Adverse events not reported.	111 patients completed the baseline tests, and 40 SOA and 36 GET participants ($N = 76$) took part for at least 3 weeks.	No support given to exercise post-discharge. One third of participants in each group dropped out due to discharge.	GET may be a beneficial adjunctive treatment in inpatients with major depression.
Buschert et al., (2019)	Depression (BDI); Hamilton Depression Rating 7-item scale; short-term verbal memory; verbal learning; executive function	Severity of depression improved significantly over time; there was no significant time by group effect.	There was a significant time × group interaction on attention and short-term memory; with greater improvements following exercise ($p = 0.01$, $\eta 2 = 0.22$; $p = 0.03$, $\eta 2 = 0.26$ respectively). There was no significant interaction effect on executive function.	Adverse events not reported.	Thirty participants completed the intervention (Exercise = 15, OT = 15). The exercise group completed on average 10 training sessions over 3–5 weeks.	No support given to exercise post-discharge. Two patients with early discharge had to be excluded from the study.	Physical exercise may improve cognition in depressive patients.
Roy et al., (2018)	Depression (HDRS-17); visual analogue mood scale (VAMS); 22-item Depression Somatic Symptom Scale (DSSS)	The estimated mean change within the group in HDRS was 4.30 ($p < 0.001$) in the exercise group as compared to 2.70 ($p < 0.001$) in control group, the difference was significant. No significant reduction in VAMS score and mean somatic symptom score between groups.		Adverse events not reported.	20 subjects in each group completed the study due to 1 dropout and 2 early discharges in study group and 3 early discharges in the control group.	No support given to exercise post-discharge. Two patients in the exercise group and three in the control group with early discharge had to be excluded from the study.	Add-on aerobic exercise is associated with greater improvements in depressive symptoms of women with depression.
Hanssen et al., 2018a	Depression (BDI); Arterial Stiffness (Augmentation index (AIx), Augmentations index corrected for 75 beats per minute (AIx@75), pulse wave velocity (PWV); VO2max	HIIT showed an 85% beneficial effect in lowering BDI-II scores compared to MICT (SMD −0.45 [90% CI − 0.85; −0.04]).	There was no change in AIx following HIIT ($p = 0.53$) and no between-group differences (SMD−0.29 [90% CI − 0.90; 0.32]). Reduction of AIx@75 was greater following MICT (SMD = 0.61) compared to HIIT (SMD = 0.08). No change in aortic pulse wave velocity.	Adverse events not reported.	A drop-out rate of 25% and 32% was observed in HIIT and MICT groups respectively. Dropout was noted to be the result of a lack of motivation.	No support given to exercise post-discharge.	Both HIIT and MICT led to a reduction in depression. HIIT was more effective in lowering depression severity, yet MICT was more effective in lowering peripheral arterial stiffness.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Hanssen et al., 2018b	Depression (BDI); Retinal vessel diameters; VO2 max	There was a significant reduction in depression severity in both groups (HIIT SMD:1.30, MICT SMD: 1.40) but no between-group differences.	There was an increase in arterio-venous diameter in both groups. There was a significant decrease in arterio-venous diameter ratio (AVR) (SMD:0.5 [95% CI-0.04; 0.96], $p = 0.035$) and no between-group difference. There was no change in retinal venules following HIIT, and a slight increase following MICT (SMD−0.15 [90% CI 0.47; 0.18]). No change in VO2max.	No adverse events.	A drop-out rate of 25% and 31% was observed in HIIT and MICT groups respectively. Dropout was noted to be the result of a lack of motivation.	No support given to exercise post-discharge.	Exercise may improve retinal vessel diameters, which is a biomarker for CVD. Both HIIT and MICT lower depression.
Minghetti et al., (2018)	Depression (BDI); Submaximal and maximal fitness parameters including: VO2 [L min−1]; VO2 [mL min−1 kg − 1]; and HR [min−1]	There was a significant reduction in BDI scores in both groups (SIT=SMD 1.1, MICT=SMD 1.2, $p < 0.001$, $\eta p^2 = 0.70$) with no differences between groups ($p = 0.98$, $\eta p^2 = 0.001$).	Submaximal ($0.07 < d < 0.89$) and maximal ($0.05 < d < 0.85$) fitness improved following both SIT and MICT, no differences were observed between groups. A significant improvement from pre-post intervention, following both SIT and MICT was found for relative oxygen uptake ($p = 0.006$, $\eta p^2 = 0.14$). There was an improvement in average HR ($p = 0.01$, $\eta p^2 = 0.16$) in both groups.	No adverse events.	A drop-out rate of 17% and 19% was observed in SIT and MICT groups respectively. Dropout was noted to be the result of a lack of motivation.	No support given to exercise post-discharge.	SIT led to similar symptom reductions in patients with depression compared to MICT.
Gerber et al., (2018)	Depression (BDI); Exercise and Sport Motivation; VO2max; Affective Valence (single-item Feeling Scale (FS)); Perceived Fitness	Depressive symptom significantly improved in the SIT and MICT group, there was no between-group interaction.	Participants in the SIT and MICT group showed similar (positive) affective responses during and after the training sessions. Cardiorespiratory fitness, self-determined motivation and self-perceived fitness similarly improved in the SIT and CAT group.	No serious adverse events.	A drop-out rate of 17% and 19% was observed in SIT and MICT groups respectively. Dropout was noted to be the result of a lack of motivation.	No support given to exercise post-discharge.	SIT led to similar symptom reductions in patients with depression compared to MICT.
Schneider et al., (2016)	Obsessive-Compulsive Drinking Scale (OCDS); Alcohol Craving Questionnaire (ACQ); Alcohol Use Disorder Test (AUDIT)	No significant Group × Time interaction for either OCDS nor ACQ. There was a significant effect for Time for OCDS ($p < 0.001$) and ACQ ($p = 0.01$). Change in OCDS total and subscale scores was significant for both groups after four weeks ($p < 0.01$) and six months ($p < 0.001$).		Adverse events not reported.	Completion rate for post (4 weeks) was 66% for the Association Splitting group and 70% for the Exercise Therapy group. After six months, the completion rate was 46% for Association Splitting and 60% for Exercise Therapy.	No support given to exercise post-discharge. Two patients withdrew from the exercise group and one from the AS group due to discharge.	There was no advantage for Association Splitting after three treatment sessions relative to Exercise Therapy.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Legrand et al., 2016	Beck Depression Inventory (BDI-II)	Depression scores improved in both the AE and ST groups but not the control group. The effect size comparing the aerobic to the control group was large (Cohen's d=−1.39), as was the effect size comparing the aerobic to the stretching group (Cohen's d=−1.06).		Adverse events in the AE group included transient muscular/ joint soreness ($n = 3$), headache ($n = 1$), and fatigue ($n = 2$). All of these participants continued the exercise program.	One of the 14 (7.1%) participants in the AE group, and two of the 11 (18.2%) participants in the ST group failed to attend≥8 training sessions. All other AE and ST participants completed the intervention (completed≥8 sessions).	No support given to exercise post-discharge.	AE may improve depressive symptoms in inpatients with MDD.
Loh et al., 2016	SF-36; PANSS; Personal and Social Performance (PSP) Scale	Statistically significant increase in the median SF-36 scores and median PSP score, and reduction of median PANSS positive and negative symptom scores and general psychopathology in the intervention group compared to treatment-as-usual group.		Adverse events not reported.	In the intervention group, four (7.8%) participants dropped out of the study.	No support given to exercise post-discharge. One patient withdrew from the exercise group due to discharge.	Walking may improve some aspects of mental health in inpatients with schizophrenia.
Salehi et al., (2016)	Beck Depression Inventory (BDI-II); Hamilton Depression Rating Scale	BDI and HDRS scores decreased significantly from pre-to post-assessment in all groups. There was a significant Time by Group Interaction; BDI and HRRS scores were significantly lower in the ECT + exercise group compared to the ECT and exercise groups.	pBDNF scores increased significantly from pre-to post-assessment. There was a statistically significant Time by Group Interaction; pBDNF levels were lowest in the exercise as compared to the ECT and ECT + exercise conditions	Adverse events not reported.	All patients were compliant with the exercise.	No support given to exercise post-discharge.	ECT and exercise and their combined use may improve depressive severity in patients with MDD.
Schuch et al., (2015)	Depression (HDRS17); WHOQOL BREF QoL tool	There was a greater improvement in depression at discharge following exercise compared to control (95% CI: 6.21–1.19; $p = 0.005$).	There were significant differences favourable to exercise group at discharge on psychological (95% CI: 9.58–28.62; $p = 0.01$) and physical (95% CI: 6.3–−23.45; $p = 0.001$) domains of QoL.	There were no adverse events in the exercise group.	Two dropouts were reported in the intervention group (8%) and one in the usual care group (4%). Twenty-two exercise sessions were not performed out of 237 (9.28%).	No support given to exercise post-discharge. Exercise was given until discharge (the end of intervention time point).	Physical exercise could be an effective intervention for improving depressive symptoms and QoL.
Schuch et al., (2014) (secondary analysis of Schuch et al., 2015)	Oxidative stress (TBARS) serum levels; BDNF assay		Exercise had a significant effect on TBARS compared to control (Exercise: pre 46.15 ± 4.2, post 35.48 ± 3.7; Con: pre 46.31 ± 6.0, post 54.29 ± 5.2 μM of MDA). No group × time interaction was found ($p = 0.13$) for BDNF serum levels.	Adverse events not reported.	There were no dropouts. The participants of the exercise group performed a median of nine sessions (range 6–14).	No support given to exercise post-discharge. Exercise was given until discharge (the end of intervention time point).	Exercise decreases the TBARS serum levels of severely depressed inpatients but had no effects on BDNF.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Kerling et al., (2015)	Montgomery-Åsberg Depression Rating Scale (MADRS); BDI; glucose and lipid profiles; exercise capacity; cardiorespiratory fitness; WC	There was no significant difference between groups in improvement in MADRS (F = 2.23; $P = 0.14$) or BD (F = 0.69; $P = 0.41$). More patients in the exercise group were classified as responders (≥50% reduction in MADRS score) compared to TAU (exercise: 14/22 versus TAU: 6/20; $\chi 2 = 4.7$; $P = 0.037$).	There were greater reductions for HDL cholesterol ($P < 0.001$) and WC ($P = 0.014$) following exercise compared to TAU. There was a significantly greater improvement in resting HR, HR at 50W, maximum power output, VO2peak following exercise compared to TAU.	Adverse events not reported.	All the 22 exercise patients completed the study. The patients participated in over 90% of the possible training units.	No support given to exercise post-discharge.	Adjunctive exercise training in depressed inpatients may improve physical fitness, MetS factors, and psychological outcome.
Kerling et al., (2017) (secondary analysis of Kerling et al., 2015)	Serum BDNF immunoassay		There was a significant time x group effect for BDNF (F = 5.05; df = 1; $p = 0.03$), suggesting an increase in BDNF concentrations in the exercise group compared to TAU.	Adverse events not reported.	All the 22 exercise patients completed the study. The patients participated in over 90% of the possible training units.	No support given to exercise post-discharge.	Adjunctive exercise in depressed inpatients may have beneficial effects on BDNF.
Kerling et al., (2018) (secondary analysis of Kerling et al., 2015)	The amount of intra-abdominal adipose tissue, pericardial adipose tissue, and muscle mass		Repeated measures ANOVA revealed a significant effect of exercise training on muscle mass (F = 4.6; df = 1; $P = 0.042$).	Adverse events not reported.	All the exercise patients completed the study. On average, >90% (16.7 of 18) of the possible training units were performed.	No support given to exercise post-discharge.	Regular exercise increases muscle mass in patients with MDD.
Kahl et al., (2016) (secondary analysis of Kerling et al., 2015)	The amount of epicardial, intra-abdominal and subcutaneous adipose tissue; anthropometric variables		There was a favourable effect of exercise for epicardial adipose tissue (F = 6.39; df = 1; $P = 0.017$) and subcutaneous adipose tissue (F = 5.8; df = 1; $P = 0.023$). There was no significant treatment x time effect for intra-abdominal adipose tissue. Significant effects of the training condition were observed on weight ($P = 0.047$), BMI ($P = 0.04$), HDL ($P = 0.021$).	Adverse events not reported.	Attendance not reported.	No support given to exercise post-discharge.	Adjunctive exercise in depressed inpatients may reduce the amount of visceral tissue and ameliorate factors constituting the MetS.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Rosenbaum et al., (2015)	PTSD checklist – civilian version (PCL-C); The Depression Anxiety and Stress Scale (DASS); the International Physical Activity Questionnaire – Short Form (IPAQ-SF); Pittsburgh Sleep Quality Index (PSQI); Pittsburgh Sleep Quality Index Addendum for PTSD (PSQIA); 6-min walk test (cardiorespiratory fitness)	A significant improvement was found in the intervention group compared with the usual care group for PTSD symptoms (PCL-C score) (MD = −5.4, $P = 0.04$), depression, stress and anxiety (DASS total MD= −17.4, $P = 0.004$; depression domain MD= −7.0, $P = 0.006$, anxiety domain MD=−6.3, $P = 0.003$, stress domain MD= −4.6, $P = 0.03$), and sleep quality (PSQIA total score (MD = −2.5, $P = 0.04$ (change in PSQI score was not significant)).	The intervention group demonstrated a significantly greater reduction in body fat percentage (MD = −5.9%, $P = 0.001$), WC (MD = −3.6 cm, $P = 0.04$), time sitting (MD = −261 min, $P = 0.001$) compared with the usual care group. There were no between-group differences in body weight, BMI, cardiorespiratory fitness, resting HR, SBP, DBP, grip strength and total physical activity.	No exercise related adverse events.	Participants in the exercise group completed a mean of seven supervised exercise sessions (SD = 2, 58% mean attendance) out of 12. 58 (71%) participants completed the follow-up assessment.	Average length of stay for the in-patient programme was 3 weeks, with most patients continuing a less-intensive out-patient programme for several months.	Exercise is associated with reduced PTSD, depressive symptoms and WC and improved sleep quality in patients with PTSD.
Heissel et al., (2015)	Depression (BDI); 6-min Walk Test (SMWT); Get-Up and Go Test (TGUG) (physical functional capacity); Neurocognitive Tests; BDNF	There was a greater reduction in BDI score following exercise compared to control ($F(1, 10) = 13.102$, $p < .005$).	There were no between-group differences in SMWT, TGUG, BDNF nor cognition (Stroop task and the LPS-3 task).	Adverse events not reported.	Two patients dropped out during treatment giving a final sample of 12 patients; thus the completion rate was 85%. The remaining participants "attended in almost every session".	No support given to exercise post-discharge. Two patients dropped out as they were discharged from hospital.	Exercise may be an effective add-on for depressive symptoms in older inpatients with depression but a larger sample is needed.
Shachar-Malach et al., 2015	Depression (HAM-D; BDI); Clinical Global Impression Scale (CGI); Visual Analog Scale (mood, VAS)	According to the HAM-D, there were four (out of six) responders (>50% reduction in HAM-D score) in the aerobic group, and none in the stretching group. According to the BDI, there were two responders in the aerobic group and none in the stretching group. Improvement in the aerobic group was significantly greater than the stretching group on three scales (CGI: $p = 0.02$; BDI: $p = 0.008$; VAS: $p = 0.03$).		No adverse effects in either exercise group.	There were no dropouts.	No support given to exercise post-discharge.	AE significantly improves treatment outcome in inpatients with MDD but a larger sample is needed.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Kim et al., (2014)	BMI; WC; BP; physical fitness (handgrip, leg and muscle strength); colonic transit time (CTT)		Exercise yielded a greater decrease in SBP and total CTT compared to control ($P = 0.016$ and $P = 0.027$) and greater improvements in leg muscle strength ($P = 0.005$), cardio-respiratory endurance ($P = 0.048$), leg muscle elasticity and power output ($P = 0.015$). Changes in body weight, BMI, body fat percentage, WC, DBP, resting HR and grip strength were similar between groups.	Adverse events not reported.	The study dropout rate was 30.0%, with 23 patients in the exercise group and 19 patients in the control group completing the study (30 were allocated to each group).	No support given to exercise post-discharge.	A 12-week AE program can benefit psychiatric inpatients by increasing intestinal motility, and decreasing some CVD risk factors.
Kim et al., 2014b	Body weight; BMI; fat free mass; body percentage; blood lipid profiles; muscle mass; cardiovascular fitness; muscle and grip strength; Serum BDNF		There was a significant improvement in body weight, BMI, body fat percentage, lean body mass, WC, SBP, DBP, resting HR, leg strength, cardiovascular fitness and serum BDNF following exercise but not following control. No changes in fasting blood glucose, total cholesterol, LDL-C, HDL-C, and TG.	Adverse events not reported.	Four participants (one of exercise group and three of control group) were dropped out from the study during the experiment due to personal reasons or health reasons unrelated to the study.	No support given to exercise post-discharge. All participants had been chronically hospitalized for more than 3 years.	Exercise may improve BDNF and some physical fitness parameters in inpatients with schizophrenia but more work is needed.
Ho et al., 2014	Depression (MADRS); The Chinese Beck Depression Inventory (C-BDI); Sit-and-Reach Flexibility Test (Low back and hamstring flexibility); handgrip test	Subjects in aerobic exercise group showed a significant reduction in depressive scores (MADRS) compared to control (between-group mean difference = 10.08 ± 9.41; $P = 0.026$). The difference in C-BDI between groups was non-significant ($P = 0.103$).	The exercise group showed a significant improvement in sit-and-reach flexibility, while in control group, no significant increase in sit-and-reach distance was observed ($p = 0.02$). No changes in handgrip strength.	No adverse side effects.	12 participants (exercise group = 7, control group = 5) were unable to finish training, reasons included early discharge ($n = 11$) and change in medical condition ($n = 1$). Forty participants completed the program.	No support given to exercise post-discharge. Ten subjects were ineligible due to early discharge and a further 11 dropped out due to discharge.	AE may reduce depressive symptoms and increase flexibility among inpatients with mild to moderate depression.
Manjunath et al., (2013)	Clinical Global Impression Severity (CGIS); PANSS; Depression (HDRS); Simpson Angus Scale (SAS) (extrapyramidal side-effects)	CGIS, PANSS and HDRS scores dropped in both groups. There was an advantage for yoga over exercise in reducing CGIS and HDRS scores.	Total SAS score dropped in both groups, with no between-group difference.	Patients did not spontaneously report any exercise-related adverse event.	82 participants completed the 2-week intervention. The proportions of patients dropping out at the 2nd week were not significantly different across groups (exercise = 5; yoga = 1). At the 6th week, more patients in the exercise group had dropped out (19 vs. 9; $P < 0.04$).	The family members, who were observing their training during their inpatient stay, monitored their performance at home after discharge.	Exercise and yoga may reduce illness severity in inpatients with psychosis.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Knubben et al., (2007)	Bech-Rafaelsen Melancholy Scale (BRMS) for depression; Center for Epidemiologic Studies Depression scale (CES-D);	Reduction in depression in the exercise group was significantly larger than in the control group (BRMS: 36% v 18%; CES-D: 41% v 21%; $p = 0.01$); the proportion of patients with a clinical response (reduction in the BRMS scores by ≥ 6 points) was larger for the exercise group (65% v 22%, $p < 0.01$).		Patients reported no negative effects of exercise (muscle pain, tightness or fatigue).	Three patients (1 exercise, 2 control) dropped out for personal reasons	No support given to exercise post-discharge.	Endurance exercise may improve mood in patients with major depression in a short time.
Knapen et al., (2003)	Cardiorespiratory fitness; Maximum dynamic muscular strength; Abdominal muscle endurance		The PF group achieved average gains of 13.59% and 14.53% in the physical work capacity at 60% and 80% of HRR, compared to gains of 0.31% and 2.28% in the GPMT group. The PF group had greater gains in muscular fitness (F1.120 = 13.83, $p = 0.0003$).	Adverse events not reported.	57 and 51 participants from the PF and GPMT groups completed the fitness assessments respectively. All patients, except six who dropped out immediately after randomization, remained in the study until their discharge. Attendance was 85.42% and 84.33% in PF and GPMT groups respectively.	No support given to exercise post-discharge. 33 patients in the exercise group were discharged at week 8 and 19 prior to the 16-week end of intervention, these patients dropped out of the study.	PF may lead to greater gains in cardiorespiratory fitness compared to GPMT.
Knapen et al., (2005) (secondary analysis of Knapen et al., (2003))	Depression (BDI); The Trait Anxiety Inventory (TAI); The Rosenberg Self-Esteem Scale (RSES); The Physical Self-Perception Profile (PSPP)	Both groups showed significant improvements on the BDI, TAI and RSES, with no between group differences.	Both groups showed significant improvements on the PSPP, with no between group differences.	Adverse events not reported.	57 and 51 participants from the PF and GPMT groups completed the intervention respectively. Attendance was 85.42% and 84.33% in PF and GPMT groups respectively.	No support given to exercise post-discharge. 39 patients were discharged at week 8 and 52 prior to the 16-week end of intervention, these patients dropped out of the study.	PF and GPMT are equally effective in improving mental health and physical self-concept.
Palmer et al., (1995)	Depression (CES-D); resting pulse rate; BP; maximum strength on incline bench press; aerobic fitness; body fat	The bodybuilding group demonstrated a significant reduction in CES-D scores (t14 = 3.43, $p < 0.01$). There was no significant reduction in CES-D scores for either the circuit training group or the aerobic group.	None of the exercise programs produced significant changes in blood pressure, resting pulse, Kasch Step Test (aerobic fitness), triceps skinfolds, or incline bench press.	Adverse events not reported.	Eighty patients volunteered to participate. Of these 80, 35 left the facility before finishing the program.	No support given to exercise post-discharge. 35 patients were discharged before finishing the program.	Bodybuilding may improve depressive symptoms in patients with substance and alcohol abuse.

(*Continued*)

Table 3. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Bosscher, 1993	Zung self-rating depression scale (SDS); Self-esteem scale (SES); Hopkins symptoms checklist (HSCL) (discomfort and wellbeing); Body cathexis scale (BCS) (satisfaction with body parts)	One way analysis showed no between group differences for depression, wellbeing or self-esteem but pre-post analysis showed significantly improvement on all of these measures following running but not TAU. Two by two analysis for variance revealed a significant treatment X time effect for depression ($p = 0.07$).	One way analysis showed no between group difference for BCS score. However, pre-post analysis showed the running group significantly improved on this measure whereas the TAU group did not. Two by two analysis for variance revealed a significant treatment X time effect for body satisfaction ($p < 0.05$).	Adverse events not reported.	24 participants were recruited, 5 did not complete the study due to discharge and one due to dissatisfaction with the exercise program. Patients attended an average of 21 sessions (out of 24), with no less than 18 sessions attended. The TAU group attended an average of 20 sessions.	No support given to exercise post-discharge. Due to patient discharge, five patients did not complete the study.	Running may be an effective treatment for depression and generalized feelings of discomfort in mental health inpatients.
Sexton et al., (1989)	Symptom Checklist (SCL-90 R); State-Trait Anxiety Inventory (STAI); Depression (BDI); Brief Psychiatric Rating Scale (BPRS); Global Assessment Scale (GAS); aerobic capacity	There were no significant differences between walkers and joggers for STAI, BDI, BPRS, GAS and SCL-90 R scores at intervention completion nor at 6-month follow-up.	At completion of the program joggers had significantly greater aerobic gains than the walkers (6.5, ±4.9 vs. 2,7 ± 4.4 ml/kg/min, $P < 0.05$. Six months after completion of the program both walkers and joggers had persistent significant increases in their aerobic capacity.	Adverse events not reported.	Three joggers and one walker did not complete the hospital phase of the study. 23 walkers and 17 joggers completed the post-hospital phase. The difference in drop-out rate between walkers and joggers was significant ($P < 0.005$)	After discharge from the hospital, participantsexercised on their own until completion ofthe 8-week period, no support during this period was noted.	Jogging and walking have equal psychological effects in non-psychotic inpatients, although jogging may have greater aerobic benefits.
Martinsen et al., (1989)	Depression (MADRS; BDI); VO2max	There was a significant reduction in depression scores in both groups ($P < .001$) and no between-group difference.	There was a significantly greater increased in VO2max in the aerobic group compared to the non-aerobic group ($P < .001$).	One patient in the aerobic group had to stop due to a hiatus hernia.	Nine patients dropped out of the study, eight in the aerobic and one in the nonaerobic.	No support given to exercise post-discharge. One patient in the nonaerobic group and six in the aerobic group dropped out due to discharge.	Both aerobic and non-aerobic training may improve symptoms of depression in inpatients with depressive disorders.
Martinsen et al., 1989b	Comprehensive Psychopathological Rating Scale (CPRS) anxiety subscale; Phobic Avoidance Rating Scale (PARS); Agoraphobic Cognitions Scale (ACS); VO2 max	There was a significant reduction in anxiety in both groups ($P < .001$) and no between-group difference.	There was a significantly greater increased in VO2max in the aerobic group compared to the non-aerobic group ($P < .001$).	Three in the aerobic group withdrew from training, finding it 'too strenuous. One experienced problems with a hiatus hernia during exercise.	Nine subjects dropped out of the study, five in the aerobic and four in the non-aerobic group.	No support given to exercise post-discharge. Four patients in the nonaerobic group and one in the aerobic group dropped out due to discharge.	Both aerobic and non-aerobic training may improve symptoms of anxiety in inpatients with anxiety disorders.
Gary & Guthrie, 1972	The Jourard Body-Cathexis and Self-Cathexis Scales (self-concept); pulse; a log of drinking behaviours and sleep disturbances	There was a significantly lower level of sleep disturbances following exercise but there was no change in the number of drinking episodes noticed by staff.	Pulse rate significantly decreased in the exercise group but not in the control group. Self-cathexis scores improved significantly (mean change: exercise 0.34 ± 0.424, control −0.18 ± 0.294).	Adverse events not reported.	Attendance not reported.	No support given to exercise post-discharge.	Exercise may be a useful treatment addition for alcoholism.

Table 4. Mental health findings, additional outcomes, adherence and adverse events from included Non-RCTs.

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Durmus et al., 2020	Hamilton Depression Rating Scale (HAM-D); the Hamilton Anxiety Rating Scale (HAM-A); the Substance Craving Scale (SCS); Cytokines	Comparison of the pre- and the post- treatment performances of the two groups on the HAM-D ($p < 0.001$), the HAM-A ($p = 0.021$) and the SCS ($p = 0.001$) indicated a significant drop in the HIIT group.	A significant increase was observed in the post-treatment IGF-1 level of the HIIT as compared to control ($p = 0.003$). No differences were observed between the cortisol, IFN-γ and IL-17 levels of the HIIT and control groups.	One participant incurred muscle injury outside the treatment program, treatment related events not reported.	Five of the 11 HIIT participants completed the full program, others dropped out due to personal reasons, beta agonist therapy, non-exercise related muscle injury.	No support given to exercise post-discharge.	HIIT may be a beneficial adjunctive treatment in inpatients with OPuD.
Kern et al., (2020)	BMI; exercise dependence; perceived physical activity; eating disorder (ED) symptoms	There was a significant improvement in three out of four dimensions of the Eating Disorders Examination-Questionnaire (EDE-Q). EDE-Q global score significantly decreased ($p = 0.01$; ES = 0.6).	BMI ($p < 0.001$, ES=−0.6) and perceived health of the DUKE Health Profile scores ($p = 0.01$; ES=−0.7) significantly increased. Exercise dependence scores significantly decreased ($p = 0.01$, ES = 0.37).	The exercise program was "safely and well adapted", specificities not reported.	41 participants were recruited, 29 of whom completed all the exercise sessions and post-test evaluation.	No support given to exercise post-discharge. Several patients dropped out due to discharge.	The APA program may be a useful program for AN inpatients to improve ED symptoms and their dependence to physical activity.
Stanley et al., (2019)	Weight; WC; Waist-to-hip ratio (WHR), BP; strength and agility; HR		No statistically significant differences were found between pre and post programme weight scores, WC, WHR, BP. Significant improvements were found on strength and agility, and resting and post exercise HR. The number of baseline readings indicating hypertension reduced from 14 to 8 people.	Adverse events not reported.	Attendance not reported.	Participants discharged from the service could continue the exercise programme as outpatients, and community patients could join on referral.	The exercise programme may improve physical capacity of participants.

(*Continued*)

Table 4. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Mazyarkin et al., (2019)	Positive and Negative Symptom Rating Scale (PANSS); the Brief Psychiatric Rating Scale (BPRS); The empathy quotient scale (EQ SCALE); The general wellbeing scale (GWBS); Visual Analogue Scale (VAS) (mood); BMI; fasting blood glucose, lipid and cholesterol profiles	Improved in negative symptoms and PANSS total score, but not in positive symptoms, was observed in the exercise group over 3 months group compared with the physical activity up to 3 months and control groups (F = 3.3, df = 2.28, p = 0.049 and F = 6.6, df = 2,228, p = 0.0045 respectively). Improvement in the PANSS general score (F = 4.9, df = 228, p = 0.018) was observed in both exercise groups compared to control. An improvement in the BPRS score was found in the exercise group (independent of training time) compared to the control group (F = 4.9, df = 2.28, p = 0.018). No differences in wellbeing, mood, satisfaction, and functioning between groups.	Furthermore, physical activity did not improve any of the physiological parameters that were measured in this study.	Adverse events not reported.	Eleven patients completed exercise for up to three months and 12 patients completed exercise for up to six months.	No support given to exercise post-discharge.	Exercise may improve mental health parameters with greater gains observed when exercise is carried out for at least 3 months.
Wunram et al., (2018)	Depression Inventory for Children and Adolescents (DIKJ); Magglinger Sports Enjoyment Scale (MSES); BMI; Body fat; Cardiorespiratory fitness	Compared to TAU, Exercise and WBV showed a significant decrease in depression scores at week 26 (p = 0.037 for ergometer and p = 0.042 for WBV). Remission rates (DIKJ scores<18) amounted to 39.7% after week 6 and 66% after week 26, compared to 25% after week 26 in TAU.	In the MSES, the question "Fun" was rated statistically higher from the ergometer group compared to WBV. BMI percentiles and body fat did not decrease. Cardiorespiratory fitness (VO2 peak/kg) improved for the cycling condition but not for the WBV group.	No exercise related adverse events (AE) nor serious adverse events (SAE).	Of 64 participants included in the study, 12 dropped out (ergometer = 3, WBV = 3, control = 6; p = 0.565). The number of trainings attended did not differ significantly with 23.5 ± 2.5 (of 28) interventions for the ergometer condition and 22.1 ± 3.70 interventions (of 27) for the WBV group.	No support given to exercise post-discharge. Most patients who dropped out were released from their inpatient treatment. Motivation to continue exercising after the programme (and after discharge) was assessed. Subjects were screened for inclusion in the inpatient and day-clinic psychiatry departments, 89.1% of the total sample were inpatients, 10.9% received day-clinic treatment.	Exercise may be an effective add-on for depressive symptoms in adolescent inpatients with depression.

(*Continued*)

Table 4. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Wunram et al., (2021) (secondary analysis of Wunram et al., 2018)	BDNF; IGF-1; Depression Inventory for Children and Adolescents (DIKJ); Beck depression inventory II (BDI-II)		The two intervention groups had higher BDNF concentrations than the controls after 6-weeks intervention (Exercise $p = 0.009$, WBV $p = 0.013$). The IGF-1 concentration in controls and WBV did not change significantly, while the mean IGF-1 concentration of the Exercise-group increased ($p = 0.011$). BDNF and IGF-1 showed no influence on depression-scores.	No exercise related adverse events (AE).	Of 64 participants included in the study, 12 dropped out (ergometer = 3, WBV = 3, control = 6; $p = 0.565$).	No support given to exercise post-discharge.	Endurance and muscle strengthening trainings may influence serological BDNF and IGF-1.
Song et al., (2018)	BMI; WC; BP; physical fitness (handgrip, leg and muscle strength); colonic transit time (CTT)		There were no significant changes in body fat percentage, SBP, DBP, resting HR, WC, lean body mass, thigh circumference, grip strength and cardiorespiratory endurance between groups. Improvements in leg muscle strength ($P = 0.009$), were significantly greater following exercise, and the reduction in the total CTT and recto-sigmoid CTT was significantly better for the exercise group compared to control ($P = 0.019$ and $P = 0.041$ respectively).	Adverse events not reported.	Among the 40 participants in the exercise group, 9 left the study due to personal illness, non-participation in exercise, or a change in medication. Among the 30 participants in the control group, 9 left the study due to non-participation, discontinuation of further exercise, or for personal reasons.	No support given to exercise post-discharge.	A 12 week combined exercise programme may increase intestinal motility.
Woodward et al., (2018)	PANSS; Calgary Depression Scale (CDS); Anxiety (HAM-A); SOFAS (Social and Occupational Assessment Scale, social functioning); Hippocampal volume; physical health	At follow-up, PANSS scores ($t(16) = -16.8$, $p \leq 0.0001$) and social functioning ($t(16) = 4.4$, $p = 0.0004$) improved. A trend for improved depression scores ($t(16) = -2.0$, $p = 0.06$). No change in anxiety.	Hippocampal volume increased ($t(16) = -2.54$, $p = 0.02$). There was no change in hippocampal vascular volume, resting HR, resting BP, BMI and TG levels, VO2max or blood cholesterol.	Adverse events not reported.	Twenty-seven participants were recruited and 17 completed the programme, participants were lost due to symptom exacerbation, discharge and health issues. Adherence rates were 83% ±9.4% with 70% of participants completing the entire programme.	No support given to exercise post-discharge. Six participants lost due to discharge.	Adjunct exercise may accelerate symptom improvement in treatment-resistant psychosis patients and cause hippocampal plasticity.

(*Continued*)

Table 4. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Kerling et al., (2016)	VO2peak; VAT (ventilatory anaerobic threshold); maximum power		Compared to controls, patients with MDD had significantly lower aerobic capacity (VAT, VO2peak, Powermax). After training, there was a significant improvement in their performance. A significant difference remained only for VAT between patients with MDD and healthy controls.	Adverse events not reported.	All 25 study participants finished the training programme, and on average, 16.4 of 18 possible training units were performed.	The possibility of training in the department after dismissal from the inpatient treatment was available.	A structured exercise is a useful supplement for patients with MDD in terms of improving aerobic capacity.
Heggelund et al., (2011)	The Calgary Depression Scale for Schizophrenia (CDSS); PANSS; Cardiorespiratory fitness; lipid and cholesterol profiles; BMI; QoL	There were no improvements in positive and negative symptoms and depression severity in either group.	Following HIIT, VO2peak increased by 12%, more than the CG group ($P = 0.014$). Net mechanical efficiency of walking improved by 12% in the HIT group, more than the CG group ($p = 0.031$). HDL-cholesterol decreased by 8 ± 8% following CGP ($p = 0.044$), although there was no change following HIIT. No changes in BMI, SBP, DBP, TG, LDL, CRP, TC, and QoL in either group.	One participant in the HIIT group was excluded due to ankle pain, it is not clear whether HIIT caused the pain or whether the pain was caused by an underlying issue.	Of the 25 participants enrolled, 6 withdrew, reasons included hospital discharge, ankle pain, excluded due to<80% attendance. Patients in the HIIT ($n = 12$) and CGP ($n = 7$) groups completed 85 ± 9% and 83 ± 6% of the training sessions, respectively.	No support given to exercise post-discharge. One patient from each group was discharged from the hospital before completion.	HIIT can improve VO2peak and net mechanical efficiency of walking in inpatients with schizophrenia.
Ng et al., (2007)	Clinical Global Impression Severity (CGI-S) and Improvement (CGI-I) scales (symptom severity); Depression Anxiety Stress Scales (DASS)	At discharge, differences in CGI-S and CGI-I scores between participants and non-participants were non-significant. Scores for total DASS and its subscales were significantly lower in participants compared with non-participants ($p < 0.05$).		Adverse events not reported.	During the recruitment timeframe there were 24 admissions (for 14 patients) who regularly attended the walking group and 74 admissions (for 35 patients) who did not attend/ attended intermittently.	No support given to exercise post-discharge. Inpatients were invited to the exercise group until discharge.	Exercise may improve depression, anxiety and stress symptoms in inpatients with Bipolar disorder.
Gaul-Alacova et al., 2005	Heart rate variability; Anxiety (HAM-A); Self-Report Symptom Inventory (SCL 90)	There was a greater decrease in anxiety in the exercise group compared to control (Exercise: decrease in test values HAMA = 63.3%, SCL 90 = 61.2%; Control: HAMA = 44.3%, SCL-90 = 33.3%).	Decreased autonomic nervous system (ANS) activity was found in patients with panic disorder, mixed anxiety, and depressive disorder. There were non-significant tendencies of an increase in vagal activity values in the exercising patients with initial normal ANS efficiency.	Adverse events not reported.	Attendance not reported.	No support given to exercise post-discharge.	Exercise may lead to a greater improvement in psychiatric outcomes compared to integrative therapy, this may be due to increased ANS efficiency.

(*Continued*)

Table 4. (Continued).

Study	Outcome and Measurement Tool	Mental Health Outcomes	Additional Outcomes	Adverse Events	Attendance (completion rates and dropout)	Support post-discharge	Conclusions
Martinsen et al., 1989c	Symptom Rating Test (SRT) (general mental distress); Depression (BDI); Physical work capacity (PWC)	The SRT total scores at discharge were significantly lower ($P < 0.05$) than at admission. Follow-up scores remained significantly lowered compared with admission among patients with depressive and alcohol disorders but not in patients with anxiety or somatoform disorders. BDI scores followed the same trend.	PWC significantly increased (92.6 ± 26.6% of predicted normal values at baseline, 109.2 ± 27.2% at discharge).	Adverse events not reported.	89 participants completed the trial, 84% of them filled in the follow-up questionnaires. Forty-nine patients (66%) reported regular exercise habits at follow-up.	No support given to exercise post-discharge.	Exercise may have mental and physical health benefits in inpatient mental health care.
Tsukue & Shohohi, 1981	WC; grip strength; vital capacity; muscle strength		Mean WC decreased 2.6%. Vital capacity increased (3.064 to 3.301 litres). Grip strength increased by 9.4%, and mean back muscle strength increased 26.5%.	Adverse events not reported.	95 patients took part in the basketball sessions. Assessments were conducted every 2 months for 10-months. 25 patients completed≥3 assessments, 8 patients completed the 10-month assessment.	No support given to exercise post-discharge.	Exercise may be a useful treatment addition for alcoholism.

Table 5. Summary of Meta-analysis Findings.

		Meta-analysis			Heterogeneity	Egger's Intercept			Begg and Mazumdar rank correlation		Duval and Tweedie trim and fill
Analysis	Number of Participants (RCTs)	Point estimate of effect sizes (SMD)	95% CI	*P* value (two-tailed)	(I2%)	Intercept	t-value	*P* value (two-tailed)	Tau	*P* value (two-tailed)	SMD [95%CI] (adjusted studies)
Exercise vs Non-exercise comparators: depression	627 (15)	−0.416	−0.787 to −0.045	0.028	79.690	−2.866	1.167	0.264	−0.248	0.198	−0.188 [−0.547; 0.170] (4)
Exercise vs Non-exercise comparators: BMI	186 (5)	−0.212	−0.507 to 0.084	0.160	0	−0.194	0.066	0.952	<0.001	1.000	No Change

Haussleiter et al., 2020; Legrand & Neff, 2016; Roy et al., 2018; Schuch et al., 2015; Wunram et al., 2018) observed a significant reduction in depressive symptoms following exercise compared to non-active control. Nine trials (8 RCTs (Imboden et al., 2020; Legrand & Neff, 2016)– (Bosscher, 1993; Buschert et al., 2019; Knubben et al., 2007; Salehi et al., 2016; Shachar-Malach et al., 2015) and one semi-randomised trial (Wunram et al., 2018)) measured depression following exercise and an active control (stretching/relaxation = 6, OT = 1, passive muscular training = 1, ECT = 1), of which 3/9 studies (33.3%) (Heissel et al., 2015; Knubben et al., 2007; Shachar-Malach et al., 2015) observed a significant reduction in depressive symptoms following exercise compared to active control, although all of these trials (9/9, 100%) observed a significant reduction in depressive symptoms, following exercise, in within-group measures. Moreover, 5 RCTs compared the effects of differing exercise modalities on depression severity (Gerber et al., 2018; Hanssen, Minghetti, Faude, et al., 2018; Hanssen, Minghetti, Faude, Schmidt-Trucksäss, et al., 2018; Martinsen, Hoffart, & Ø, 1989; Minghetti et al., 2018). One RCT observed a greater reduction in depression severity following high-intensity interval training (HIIT) compared to moderate-intensity continuous training (MICT) [(Hanssen, Minghetti, Faude, et al., 2018). Three RCTs observed no difference between HIIT and MICT (Gerber et al., 2018; Hanssen, Minghetti, Faude, Schmidt-Trucksäss, et al., 2018; Minghetti et al., 2018)], and one observed no difference between aerobic and non-aerobic exercise (Martinsen, Hoffart, & Ø, 1989).

Secondary Outcomes

Two trials (one RCT (Kerling et al., 2015) and one semi-randomised trial (Wunram et al., 2018)) assessed VO2peak and BMI following exercise and non-active control conditions; both (2/2, 100%) observed significantly greater improvements in VO2peak following exercise compared to TAU but no change in BMI (0/2, 0%). One RCT (Kahl et al., 2016; Kerling et al., 2015, 2018) found a reduction in epicardial and subcutaneous adipose tissue, high-density lipoproteins (HDL), and a significant increase in muscle mass following exercise compared to TAU, and reductions in DBP, resting HR and fasting glucose in within-group measures. Three RCTs (Imboden et al., 2021; Salehi et al., 2016; Schuch et al., 2014) and one semi-randomised trial (Wunram et al., 2021) assessed brain-derived neurotrophic factor (BDNF). Plasma and serum BDNF levels increased in within group measures (4/4, 100%) and increased compared to non-active control in one study (1/2, 100%) (Wunram et al., 2021) but did not differ between exercise and active control groups (0/3, 0%) (Wunram et al., 2021) (Salehi et al., 2016). Moreover, one RCT assessed oxidative stress serum levels and observed a significant reduction in oxidative stress compared to non-active control (Schuch et al., 2014)], and one semi-randomized trial observed an increase in IGF – in within group measures (Wunram et al., 2021).

One RCT (Gerber et al., 2020; Imboden et al., 2020) found an increase in VO2max following both exercise and a stretching control group, with no changes in cortisol reactivity, blood pressure or resting HR. Three RCTs (Buschert et al., 2019; Heissel et al., 2015; Imboden et al., 2020) measured cognitive outcomes following exercise and active control (stretching or OT). Improvements in working memory were seen following exercise compared to active control in two RCTs (2/3, 66.7%) (Buschert et al., 2019; Imboden et al., 2020).

Adherence and adverse events

Eight trials reported ≥80% attendance at exercise sessions (Bosscher, 1993; Heissel et al., 2015; Kerling et al., 2018; Legrand & Neff, 2016; Salehi et al., 2016; Schuch et al., 2015; Wunram et al., 2018) (Kerling et al., 2017). Attendance did not differ from control conditions (Bosscher, 1993; Legrand & Neff, 2016; Wunram et al., 2018). Eleven trials (57.9%) did not report attendance (Shachar-Malach et al., 2015) (Buschert et al., 2019; Hanssen, Minghetti, Faude, et al., 2018; Haussleiter et al., 2020; Martinsen, Hoffart, & Ø, 1989; Minghetti et al., 2018; Roy et al., 2018).

Nineteen trials reported the rate of participant drop-out. No drop-outs were reported in four trials (Kerling et al., 2015, 2018; Salehi et al., 2016; Shachar-Malach et al., 2015) (Kerling et al., 2017). In 15 trials, the dropout rate for the exercise group was ≤27% (CWH et al., 2014; Gerber et al., 2020; Heissel et al., 2015; Imboden et al., 2020; Legrand & Neff, 2016) (Martinsen, Hoffart, & Ø, 1989; Minghetti et al., 2018) (Hanssen, Minghetti, Faude, et al., 2018; Schuch et al., 2015) (Wunram et al., 2018), and did not differ from control conditions (Gerber et al., 2018, 2020; Imboden et al., 2020; Legrand & Neff, 2016) (Bosscher, 1993; Hanssen, Minghetti, Faude, et al., 2018; Minghetti et al., 2018; Schuch et al., 2015) (Wunram et al., 2018).

Nine trials observed no exercise-related AEs (Gerber et al., 2020; Imboden et al., 2020; Shachar-Malach et al., 2015) (Minghetti et al., 2018; Schuch et al., 2015; Wunram et al., 2018). Two trials observed AEs including transient muscular/joint soreness, headache, fatigue and hiatus hernia (Legrand & Neff, 2016; Martinsen, Hoffart, & Ø, 1989). Nine trials (45%) did not report on AEs (Bosscher, 1993; Heissel et al., 2015; Kahl et al., 2016; Kerling et al., 2018) (Buschert et al., 2019; Haussleiter et al., 2020; Kerling et al., 2015; Roy et al., 2018; Salehi et al., 2016) (Kerling et al., 2017).

Schizophrenia spectrum disorders

Eight exercise trials (across nine papers) were conducted in inpatients with schizophrenia-spectrum disorders (201 = exercise and 164 = control groups) (Mazyarkin et al., 2019)– (Shimada et al., 2019) including five RCTs (Jae et al., 2014; Manjunath et al., 2013)– (Shimada et al., 2019), two NRCTs (Heggelund et al., 2011; Mazyarkin et al., 2019) and one pre-post-test (Woodward et al., 2018).

Primary Outcomes

Five out of seven trials (71.4%) observed an improvement in Positive and Negative Syndrome Scale scores (PANSS) PANSS total scores in within-group measures.

Two RCTs (Shimada et al., 2019; SY et al., 2016) and one NRCT (Mazyarkin et al., 2019) observed a significant improvement in PANSS total score (Mazyarkin et al., 2019; Shimada et al., 2019), negative symptoms (Mazyarkin et al., 2019; Shimada et al., 2019; SY et al., 2016), positive symptoms (Shimada et al.,

2019; SY et al., 2016) and PANSS general score (Mazyarkin et al., 2019; SY et al., 2016) compared to non-active control. One RCTs (Kurebayashi et al., 2022) observed no between-group changes in PANSS scores compared to non-active control. Moreover, significant increases in cognitive domains (including attention, executive function, verbal fluency and processing speed) (Kurebayashi et al., 2022; Shimada et al., 2019) and global functioning (SY et al., 2016) were found following exercise compared to non-active control, whereas one RCT (Shimada et al., 2019, 2020) observed within group increases in global functioning but no change compared to non-active control.

One RCTs (Manjunath et al., 2013) and one NRCT (Heggelund et al., 2011) measured PANSS scores following exercise and active control conditions (yoga = 1, computer games = 1). One RCT observed a significant decrease in PANSS total score, PANSS general score, and depressive symptoms following both exercise and yoga, and a greater decrease in depressive symptoms following yoga compared to exercise (Manjunath et al., 2013). One NRCT (Heggelund et al., 2011) observed no change in either PANSS scores following exercise or computer games, and no change in depression severity following both exercise and computer games training.

One pre-post-test (Woodward et al., 2018) observed a significant decrease in PANSS total scores and an improvement in social functioning following exercise but no change in depression severity nor anxiety.

Secondary Outcomes

One NRCT found no change in BMI, fasting blood glucose and lipid profiles following up to 6 months of aerobic and resistance exercise versus a non-active control group (Mazyarkin et al., 2019)

Two controlled trials (one RCT (Jae et al., 2014) and one NRCT (Heggelund et al., 2011)) assessed a range of physical health measures following exercise and active control. One RCT observed improvements in BDNF, BMI, SBP, DBP and resting HR following exercise but not following stretching, dancing and recreation activities, and no change in HDL (Jae et al., 2014). One NRCT observed a significantly greater increase in VO2peak following HIIT compared to computer games training, a decrease in HDL following computer training but not following HIIT, and no change in BMI (Heggelund et al., 2011). Both trials (Heggelund et al., 2011; Jae et al., 2014) observed no change in total cholesterol, triglycerides, LDL, fasting blood glucose and total glycerol.

One pre-post-test (Woodward et al., 2018) observed no change in VO2max, BMI, total cholesterol, triglycerides, blood pressure and resting HR following aerobic and non-aerobic exercise. Increases in hippocampal total volume were observed.

Adherence and adverse events

Attendance to exercise sessions was reported in 2/8 trials (Heggelund et al., 2011; Woodward et al., 2018). Both observed ≥80% attendance, and attendance did not differ from control (Heggelund et al., 2011). Across seven trials the dropout rate for the exercise group ranged from 0% to 37% and did not differ from control (Heggelund et al., 2011; Jae et al., 2014; Manjunath et al., 2013) (Shimada et al., 2020). The occurrence of AEs was measured in 4/8 trials of which three experienced no exercise-related AEs (Kurebayashi et al., 2022; Manjunath et al., 2013; Shimada et al., 2020) and another experienced one account of ankle pain (Heggelund et al., 2011).

Substance and Alcohol Use Disorders

Six exercise trials were conducted in inpatients with SUD and/or AUD (144 = exercise and 63 = control groups) (Schneider et al., 2016) (McCartney et al., 2021) including four RCTs (Gary & Guthrie, 1972; McCartney et al., 2021; Palmer et al., 1995; Schneider et al., 2016), a NRCT (Dürmüş et al., 2020) and a pre-post-test (Tsukue & Shohoji, 1981).

Primary Outcomes

One RCT (Gary & Guthrie, 1972) assessed the number of drinking episodes and sleep outcomes in inpatients with AUD, following exercise and non-active control. One NRCT (Dürmüş et al., 2020) assessed substance craving, depression severity and anxiety in inpatients with opioid use disorder following exercise and non-active control. There was a significant reduction in sleep disturbances, substance craving scores, depression and anxiety following exercise compared to non-active control, and no change in the number of drinking episodes. One RCT (Schneider et al., 2016) observed a significant reduction in alcohol craving following exercise, in inpatients with AUD, but no difference versus association splitting therapy. In another RCT a significant reduction in depression was seen following bodybuilding but not following aerobic exercise nor circuit training (Palmer et al., 1995). Moreover, one RCT (McCartney et al., 2021) assessed sleep outcomes during inpatient admission for cannabis withdrawal and observed increased sleep duration and decreased wake bout following cycling compared to stretching.

Secondary Outcomes

One RCT found a significant increase in insulin-like growth factor (IGF-1) (Dürmüş et al., 2020), and a decrease in resting HR in exercise versus a non-active control (Gary & Guthrie, 1972).

Adherence and adverse events

Two trials reported attendance at individual exercise sessions (Dürmüş et al., 2020), 45% of participants attended all sessions in one trial[(Dürmüş et al., 2020) and in another 3% of sessions were missed (McCartney et al., 2021). Four trials provided participant drop-out, which ranged from 0% to 55%. The occurrence of exercise-related AEs was not reported.

Anxiety Disorders

Two exercise trials were conducted in inpatients with anxiety disorders (106 = exercise and 16 = control groups) including one RCT, which allocated patients to jogging or non-aerobic exercise (Martinsen, Hoffart, & Solberg, 1989), and one controlled trial comparing cycling combined with psychotherapy to psychotherapy alone (Gaul-Aláčová et al., 2005). Both trials observed a significant reduction in anxiety symptoms (2/2, 100%) and there was a greater decrease in anxiety symptoms following exercise compared to psychotherapy alone (Gaul-

Aláčová et al., 2005). Changes in VO2max were measured in one RCT with significant improvements following jogging but not following non-aerobic exercise (Martinsen, Hoffart, & Solberg, 1989). One trial gave details regarding the rate of participant drop-out and AE's (Martinsen, Hoffart, & Solberg, 1989). The drop-out rate was 11% and one participant experienced problems with a hiatus hernia.

Bipolar Disorder

One retrospective cohort study assessed the effects of a daily walking programme in people with bipolar disorder (Ng et al., 2007). Fourteen inpatients regularly attended the walking group until discharge whereas 35 patients did not engage. At discharge, people who regularly attended the walking group during hospitalization showed significantly lower Depression Anxiety Stress Scale (DASS) scores compared to those who did not engage. AEs were not reported.

Ptsd

A two-armed RCT assessed the effects of a 12-week exercise programme in inpatients with PTSD (Rosenbaum et al., 2015). Thirty-nine participants received the exercise programme and 42 received usual care. Following exercise, there was a significant improvement in PTSD symptoms, depression, stress, anxiety, sleep quality and body fat percentage compared to usual care. Mean attendance at supervised sessions was 58% and no exercise-related AEs were observed.

Eating Disorders

One pre-post-test intervention assessed the effects of an 8-week physical activity program, designed to reduce dependence towards unhealthy physical activity patterns, in 41 female adolescent inpatients with Anorexia Nervosa (Kern et al., 2020). A significant reduction in exercise dependence scores and eating disorders symptomology scores (eating, body shape, weight) were noted. BMI increased from pre-post-test; thus, exercise did not impede weight gain. The exercise programme was reported to be "safely and well adapted", and 70.7% of patients completed the programme.

Combined Mental Illness Samples

Eight exercise trials (across nine papers) summarised the effects of exercise in pooled inpatient samples including a range of mental health diagnoses (e.g., schizophrenia, depression, bipolar disorder, anxiety, personality disorders and SUDs) (414 = exercise and 177 = control groups). This included five RCTs (Kim et al., 2014; Knapen et al., 2003, 2005; Lan et al., 2022; Philippot et al., 2022; Sexton et al., 1989), one NRCT (Song et al., 2018) and two pre-post-tests (Martinsen, Sandvik, et al., 1989; Stanley et al., 2019).

Primary Outcomes

Four RCTs measured anxiety and depression. One RCT (Lan et al., 2022) observed a significant decrease in depression and anxiety compared to non-active control, and one RCT (Philippot et al., 2022) observed a significant decrease in depression but no change in anxiety following exercise compared to relaxation. Moreover, two RCTs observed a significant decrease in within-group measures (2/2, 100%) but no differences in magnitude of change between exercise and an active control conducting progressive sports and relaxation (Knapen et al., 2005), and between walking and jogging (Sexton et al., 1989). There was a significant decrease in the Brief Psychiatric Rating Scale (BPRS) (Sexton et al., 1989) and mental health symptom rating scores (Martinsen, Sandvik, et al., 1989) in within-group measures.

Secondary Outcomes

One RCT (Kim et al., 2014) and one NRCT (Song et al., 2018) measured changes in BMI, blood pressure, resting HR and gastrointestinal motility (colonic transit time (CTT)). One trial (Kim et al., 2014)(1/2, 50%) observed a greater decrease in SBP following exercise, compared to non-active control, and both trials observed within-group decreases in DBP and resting HR. A greater reducation in total CTT was observed following exercise compared to non-active control in both trials (2/2, 100%).

Two RCTs (Knapen et al., 2003; Philippot et al., 2022) measured cardiorespiratory fitness by means of physical work capacity (Knapen et al., 2003) and VO2max (Philippot et al., 2022) and observed significantly greater gains in cardiorespiratory fitness following exercise compared to active controls conducting progressive sports and relaxation. One RCT (Sexton et al., 1989) measured cardiorespiratory fitness across two exercise groups. Significantly greater gains in VO2max were observed following jogging compared to walking. One pre-post-test (Stanley et al., 2019) observed a decrease in resting HR following exercise.

Adherence and adverse events

Two trials reported attendance (Knapen et al., 2003, 2005; Philippot et al., 2022), reporting 76–85.42% attendance in exercise, and no difference in attendance between exercise and control groups. Seven trials reported participant drop-out, which ranged from 16% to 44% and did not differ from control conditions (Kim et al., 2014; Philippot et al., 2022; Song et al., 2018). In a two-armed exercise intervention, a significantly higher drop-out rate was observed following jogging (39%) compared to walking (8%) (Sexton et al., 1989). Exercise-related AEs were not reported. A summary of meta-analysis findings are presented in Table 5.

Support to Exercise Post-Discharge

Forty-one trials (87.2%) gave no support to exercise post-discharge or did not report support post-discharge (Imboden et al., 2020) (Imboden et al., 2021; Schuch et al., 2015) (Kerling et al., 2015, 2017) (Kurebayashi et al., 2022; Woodward et al., 2018) (Gaul-Aláčová et al., 2005; McCartney et al., 2021) (Knapen et al., 2003; Martinsen, Sandvik, et al., 1989) (Philippot et al., 2022), and in several instances, participants with early discharge were lost from the study as they were unable to continue participating in exercise or control conditions (Bosscher, 1993; Buschert et al., 2019; CWH et al., 2014; Gerber et al., 2020; Haussleiter et al., 2020; Heissel et al., 2015; Imboden et al., 2020; Martinsen,

Hoffart, & Ø, 1989) (Heggelund et al., 2011; Kern et al., 2020; Knapen et al., 2003, 2005; Kurebayashi et al., 2022; Martinsen, Hoffart, & Solberg, 1989; Palmer et al., 1995; Philippot et al., 2022; Schneider et al., 2016; SY et al., 2016; Woodward et al., 2018; Wunram et al., 2018).

Five trials (10.6%) offered exercise post-discharge. In two trials, participants could continue with the exercise regime as outpatients (Kerling et al., 2016; Stanley et al., 2019), in one trial patients continued to exercise on their own post-discharge until completion of the 8-week intervention period with no support (Sexton et al., 1989), and in another trial, patients exercised at home with supervision of family members after discharge (Manjunath et al., 2013). In another trial, patients continued a less-intensive out-patient programme for several months post-discharge and home-based exercise sessions (Rosenbaum et al., 2015).

Feedback towards Exercise

Participant feedback was provided in four trials (8.5% of trials) (Kern et al., 2020; Martinsen, Sandvik, et al., 1989; Philippot et al., 2022; Wunram et al., 2018). In one trial (Wunram et al., 2018), ergometer training was rated as statistically more "fun" compared to an active control group conducting passive muscular training, and the item "would recommend it to others" was rated higher for the exercise group. In another trial (Martinsen, Sandvik, et al., 1989), patients were asked to rank the therapeutic elements of their inpatient treatment they found most useful. Patients retrospectively ranked physical exercise as the most important therapeutic element, followed by psychotherapy, education, and community meetings. In a further trial (Kern et al., 2020), follow-up qualitative interviews revealed patients appreciated the exercise programme, and perceived it as enjoyable and useful for muscular health and relaxation. In one trial (Philippot et al., 2022) enjoyment was rated using the VAS hedonia index, and did not differ between exercise and relaxation groups (exercise: 7.2 ± 1.9, control: 7.6 ± 1.5 on a 10-point scale).

Length of Hospital Stay

Across all patient groups, five trials reported mean length of inpatient stay and observed no significant difference between exercise and control groups (Knubben et al., 2007; Ng et al., 2007; Schuch et al., 2014, 2015; Shachar-Malach et al., 2015; Wunram et al., 2018).

Discussion

There has been a growing interest in the provision of exercise interventions among inpatient mental health services to assist the holistic recovery of people with mental illness, endorsed by numerous international and national guidelines (Lambert et al., 2017; National Institute for Health and Care Excellence NICE, 2014). The current systematic review is the most comprehensive examination of exercise interventions in inpatient psychiatry and seeks to understand and synthesise characteristics and outcomes of trials that may aid clinicians in the future translation of research into clinical practice.

Across all mental health diagnoses, exercise resulted in a moderate reduction in depression severity compared to non-exercise comparators but did not change BMI. In people hospitalised for MDD, the majority of trials (6/7, 85.7%) observed a reduction in depressive symptoms following exercise compared to non-active control, and evidence from a small number of trials suggested gains in cardiorespiratory fitness, adipose tissue, muscle mass, oxidative stress and cognition. In people hospitalised for schizophrenia spectrum disorders, 3/4 trials (75%) observed an improvement in psychiatric symptoms compared to non-active control, and there was a suggestion that exercise may improve depression and social functioning. In people receiving inpatient treatment for SUDs and AUDs, there was limited evidence that exercise may reduce craving scores, anxiety, depression and sleep disturbances. In anxiety disorders, evidence from two trials suggested a role of exercise in ameliorating anxiety. Exercise significantly reduced depression, stress and anxiety in those with bipolar disorder and PTSD and led to improvements in PTSD symptoms and sleep quality. In anorexia nervosa, exercise led to a significant reduction in exercise dependence scores and eating disorders symptomology scores but did not impede weight gain. In all categories of mental illness, there was inconclusive evidence regarding the effects of exercise on resting HR, blood pressure and lipid profiles. These narrative findings were hampered by variability of outcome measurement tools, differences in exercise regimes and population characteristics, and small sample size. Taken together, our findings suggest exercise to be a viable and effective intervention to improve some aspects of mental and physical health in inpatient mental health settings, although more research is needed to establish the full range of benefits.

Exercise interventions appear to be safe when offered in inpatient mental health settings. Eighteen intervention studies (38.3%) reported on AEs, of which 14 studies (77.8%) reported no AEs. Based on current evidence, exercise appeared to be safe in people with depression, schizophrenia, anxiety disorders, PTSD, and anorexia nervosa which confirms overall findings regarding the safety of exercise interventions in previous research in mental health outpatient and community services (Stanton & Happell, 2014b). Caveats to this are that safety was only observed from a single inpatient trial in PTSD and anorexia nervosa and safety was not reported at all in samples with SUDs and AUDs, or bipolar disorder. Further research on safety in these conditions is required, particularly as previous qualitative research in bipolar disorder has highlighted the potential for exercise to accelerate activation during mania (Malhi & Byrow, 2016; Wright et al., 2012). Interestingly, all studies that observed AE's conducted jogging or treadmill-based exercise whereas the majority of those studies that did not observe AEs conducted bicycle-based exercise regimes, or bicycle regimes combined with other exercise modalities ($n = 8$, 57%) suggesting that bicycle-based exercise may be safer than treadmill exercise. There did not appear to be any association between exercise intensity and occurrence of AEs.

To add, 17 trials gave data regarding attendance, with the majority (12/17 trials, 70.6%) reporting≥80% attendance to exercise sessions. Attendance did not appear to differ from active control groups, suggesting that non-attendance may be attributable to the mental health symptoms of the

participants, or other external factors, rather than the exercise per se. All five trials that reported lower attendance recruited patients subject to an acute inpatient stay, with mean length of stay ranging from 19 days to 6 weeks (Dürmüş et al., 2020; Imboden et al., 2020; Ng et al., 2007; Philippot et al., 2022; Rosenbaum et al., 2015). It may be that patient discharge in acute settings may be a contributing factor to declining attendance.

The rate of participant dropout was reported in 40 trials and was noted to be <30% in 36 trials. Similar to attendance, in those trials that reported highest dropout (44–55%) (Dürmüş et al., 2020; Knapen et al., 2003; Palmer et al., 1995), patients were subject to an acute inpatient stay and multiple instances of participant dropout were directly attributable to patient discharge. Six trials compared the rate of dropout between differing exercise modalities (Gerber et al., 2018; Hanssen, Minghetti, Faude, et al., 2018; Hanssen, Minghetti, Faude, Schmidt-Trucksäss, et al., 2018; Martinsen, Hoffart, & Ø, 1989; Minghetti et al., 2018; Sexton et al., 1989). Additionally, MICT and jogging were found to have a higher rate of dropout when compared to non-aerobic exercise, HIIT and walking (Gerber et al., 2018; Hanssen, Minghetti, Faude, et al., 2018; Hanssen, Minghetti, Faude, Schmidt-Trucksäss, et al., 2018; Martinsen, Hoffart, & Ø, 1989; Minghetti et al., 2018; Sexton et al., 1989).

Several controlled trials made comparisons between differing exercise regimes and associated therapeutic benefit. In one study, greater gains in PANSS scores were seen when exercise was conducted for 3–6 months compared to up to 3 months (Mazyarkin et al., 2019), suggesting that continuous long-term participation in exercise is needed before all benefits of exercise can be observed. When comparing aerobic and non-aerobic exercise, two trials observed greater gains in VO2max following aerobic exercise regimes (Martinsen, Hoffart, & Solberg, 1989; Martinsen, Hoffart, & Ø, 1989). Addressing comparisons between high- and low-intensity regimes, one RCT reported that HIIT may have a beneficial effect in lowering depression severity compared to MICT (Hanssen, Minghetti, Faude, et al., 2018) whereas three RCTs observed similar reductions in depression severity in both HIIT and MICT groups (Gerber et al., 2018; Hanssen, Minghetti, Faude, Schmidt-Trucksäss, et al., 2018; Minghetti et al., 2018). Another RCT observed greater gains in cardiorespiratory fitness following jogging compared to walking, but no difference in mental health parameters (Sexton et al., 1989). Research across community and inpatient settings has demonstrated greater improvements in depression severity following HIIT compared to MICT (Korman et al., 2020; Martland et al., 2020), however more high-quality research with a larger sample is needed before optimal modalities and intensities can be observed in mental health inpatients.

Overall, risk of bias was high; only 34 trials (72.3%) followed an RCT design of which only 45% blinded assessors. Moreover, only 8.5% of trials gave details regarding feedback or patient enjoyment which is a key factor for autonomous motivation to sustain exercise in people with SMI (Vancampfort, Stubbs, et al., 2015). Despite an expansion in the number of exercise interventions carried out in inpatient mental health settings in recent years, and indeed since the 2014 systematic review (Stanton & Happell, 2014a), more high-quality RCTs, with thorough reporting of safety, attendance, adherence and enjoyment, are needed to determine optimal parameters and aid translation into clinical practice. Consulting participants regarding their beliefs about the acceptability of exercise interventions underpins principles of "co-design" which is in turn associated with increased engagement in programmes. To this end, co-design in exercise intervention development should be explored in future (Wheeler et al., 2018).

Moreover, only 10.6% of studies offered patients the opportunity to continue the exercise regime post discharge, with varying levels of support. Future research should address the best ways to support exercise in both the inpatient environment and its continuation once discharged. Another avenue to explore in future work is the influence of exercise participation on length of hospital stay. Only five studies investigated differences in hospital stay and found no change, more research is needed before conclusions can be drawn.

Despite encouraging findings, it is important to note several limitations. Firstly, the heterogeneity in terms of study designs, outcomes, exercise regimes and population characteristics, and small sample size coupled with poor quality of many included studies and limited number of studies for each mental health diagnoses limits conclusive recommendations. A more thorough description of safety, adherence and enjoyment must be provided to draw definitive conclusions concerning the feasibility and acceptability of differing exercise regimes. Finally, we found only one study involving adolescents/youth inpatients, despite international endorsement of the importance of early intervention, highlighting the need for more work in this inpatient group (collab NICE guideline collab, year 2013year) (Shiers & Curtis, 2014).

Conclusion

Exercise regimes appear to be safe, viable and effective as therapeutic interventions in inpatient mental health settings. Evidence suggests exercise may improve depression, psychiatric symptoms, anxiety, cardiorespiratory fitness, cognition and muscle mass in inpatients with a range of mental health disorders. Additionally, exercise interventions appear to have good attendance and safety, with the caveat that no definitive conclusions can be drawn about effectiveness or safety of exercise in people with acute mania currently. The evidence base has expanded significantly in support of exercise interventions within inpatients settings over recent years though more research and thorough reporting is needed to determine optimal parameters and aid translation into clinical practice.

Declarations

Ethics approval and consent to participate – Not applicable

Consent for publication – Not applicable

Availability of data and material – All data analysed during this study is included in this published article and its supplementary information files.

Authors' contributions

RM and BS designed the systematic review and search criteria. RM and BS determined study eligibility. RM and NK assessed study quality. Data extraction was performed by RM. The paper was drafted by RM and revised by BS. All authors contributed to protocol development and read and approved the final manuscript.

Disclosure statement

BS has received Honoria from ASICS. BS and JF have consulted for Parachutebh, for an unrelated project.

Funding

This paper represents independent research funded by the National Institute for Health Research (NIHR) Biomedical Research Centre at South London and Maudsley NHS Foundation Trust and King's College London. RM is supported by a PhD studentship from the NIHR Biomedical Research Centre at South London and Maudsley NHS Foundation Trust and King's College London. BS is supported by a Clinical Lectureship (ICA-CL-2017-03-001) jointly funded by Health Education England (HEE) and the National Institute for Health Research (NIHR). The views expressed are those of the authors and not necessarily those of the NHS, the NIHR or the Department of Health and Social Care. The funding partners had no involvement in the study at any stage, nor did they influence the decision to publish. JF is supported by a University of Manchester Presidential Fellowship (P123958) and a UK Research and Innovation Future Leaders Fellowship (MR/T021780/1).

ORCID

Rebecca Martland http://orcid.org/0000-0002-4080-0171
Nicole Korman http://orcid.org/0000-0003-1414-1050
Joseph Firth http://orcid.org/0000-0002-0618-2752
Brendon Stubbs http://orcid.org/0000-0001-7387-3791

References

Barbui, C., Biancosino, B., Esposito, E., Marmai, L., Donà, S., & Grassi, L. (2007). Factors associated with antipsychotic dosing in psychiatric inpatients: A prospective study. *International Clinical Psychopharmacology*, *22*(4), 221–225. https://doi.org/10.1097/YIC.0b013e3281084ea8

Bosscher, R. J. (1993). Running and mixed physical exercises with depressed psychiatric patients. *International Journal of Sport Psychology*, *24*(2), 170–184.

Buschert, V., Prochazka, D., Bartl, H., Diemer, J., Malchow, B., Zwanzger, P., & Brunnauer, A. (2019). Effects of physical activity on cognitive performance: A controlled clinical study in depressive patients. *European Archives of Psychiatry and Clinical Neuroscience*, *269*(5), 555–563. https://doi.org/10.1007/s00406-018-0916-0

Correll, C. U., Detraux, J., & De Lepeleire, J., De Hert M. (2015). Effects of antipsychotics, antidepressants and mood stabilizers on risk for physical diseases in people with schizophrenia, depression and bipolar disorder. *World Psychiatry*, *14*(2), 119–136. https://doi.org/10.1002/wps.20204

Correll, C. U., Solmi, M., & Veronese, N., Bortolato B, Rosson S, Santonastaso P, Thapa-Chhetri N, Fornaro M, Gallicchio D, Collantoni E, Pigato G. (2017). Prevalence, incidence and mortality from cardiovascular disease in patients with pooled and specific severe mental illness: A large-scale meta-analysis of 3,211,768 patients and 113,383,368 controls. *World Psychiatry*, *16*(2), 163–180. https://doi.org/10.1002/wps.20420

CWH, H., Chan, S. C., Wong, J. S., Cheung, W. T., Chung, D. W. S., & Lau, T. F. O. (2014). Effect of Aerobic Exercise Training on Chinese Population with Mild to Moderate Depression in Hong Kong. *Rehabilitation Research and Practice*, *2014*. https://doi.org/10.1155/2014/627376

Czosnek, L., Lederman, O., Cormie, P., Zopf, E., Stubbs, B., & Rosenbaum, S. (2019). Health benefits, safety and cost of physical activity interventions for mental health conditions: A meta-review to inform translation efforts. *Mental Health and Physical Activity*, *16*, 140–151. https://doi.org/10.1016/j.mhpa.2018.11.001

De Hert, M., Dekker, J. M., Wood, D., Kahl, K. G., Holt, R. I. G., & Möller, H. -J. (2009). Cardiovascular disease and diabetes in people with severe mental illness position statement from the European Psychiatric Association (EPA), supported by the European Association for the Study of Diabetes (EASD) and the European Society of Cardiology. *European Psychiatry*, *24*, 412–424. https://doi.org/10.1016/j.eurpsy.2009.01.005

Denollet, J., Maas, K., Knottnerus, A., Keyzer, J. J., & Pop, V. J. (2009). Anxiety predicted premature all-cause and cardiovascular death in a 10-year follow-up of middle-aged women. *Journal of Clinical Epidemiology*, *62*(4), 452–456. https://doi.org/10.1016/j.jclinepi.2008.08.006

Dürmüş, P. T., Vardar, M. E., & Kaya, O., Tayfur P, Süt N, Vardar SA. (2020). Evaluation of the Effects of High Intensity Interval Training on Cytokine Levels and Clinical Course in Treatment of Opioid Use Disorder. *Turk Psikiyatr Derg*, *31*(3), 151–158.

Edward, C., Goodwind, G. M., & Fazel, S. (2014). Risks of all-cause and suicide mortality in mental disorders: A meta-review. *World Psychiatry*, *13*(2), 153–160. https://doi.org/10.1002/wps.20128

Emdin, C. A., Odutayo, A., Wong, C. X., Tran, J., Hsiao, A. J., & Hunn, B. H. M. (2016). Meta-Analysis of Anxiety as a Risk Factor for Cardiovascular Disease. *The American Journal of Cardiology*, *118*(4), 511–519. https://doi.org/10.1016/j.amjcard.2016.05.041

Firth, J., Cotter, J., Elliott, R., French, P., & Yung, A. R. (2015). A systematic review and meta-Analysis of exercise interventions in schizophrenia patients. *Psychological Medicine*, *45*(7), 1343–1361. https://doi.org/10.1017/S0033291714003110

Firth, J., Rosenbaum, S., Stubbs, B., Gorczynski, P., Yung, A. R., & Vancampfort, D. (2016). Motivating factors and barriers towards exercise in severe mental illness: A systematic review and meta-analysis. *Psychological Medicine*, *46*(14), 2869–2881. https://doi.org/10.1017/S0033291716001732

Gale, C. R., Batty, G. D., & Osborn, D. P. J., Tynelius P, Rasmussen F. (2014). Mental disorders across the adult life course and future coronary heart disease: Evidence for general susceptibility. *Circulation*, *129*(2), 186–193. https://doi.org/10.1161/CIRCULATIONAHA.113.002065

Gary, V., & Guthrie, D. (1972). The effect of jogging on physical fitness and self-concept in hospitalized alcoholics. *Quarterly Journal of Studies on Alcohol*, *33*(4), 1073–1078. https://doi.org/10.15288/qjsa.1972.33.1073

Gaul-Aláčová, P., Bouček, J., Stejskal, P., Kryl, M., Pastucha, P., & Pavlík, F. (2005). Assessment of the influence of exercise on heart rate variability in anxiety patients. *Neuroendocrinol Lett*, *26*(6), 713–718.

Gerber, M., Imboden, C., Beck, J., Brand, S., Colledge, F., Eckert, A., Holsboer-Trachsler, E., Pühse, U., & Hatzinger, M. (2020). Effects of Aerobic Exercise on Cortisol Stress Reactivity in Response to the Trier Social Stress Test in Inpatients with Major Depressive Disorders: A Randomized Controlled Trial. *Journal of Clinical Medicine*, *9*(5), 1419. https://doi.org/10.3390/jcm9051419

Gerber, M., Minghetti, A., Beck, J., Zahner, L., & Donath, L. (2018). Sprint Interval Training and Continuous Aerobic Exercise Training Have Similar Effects on Exercise Motivation and Affective Responses to Exercise in Patients with Major Depressive Disorders: A Randomized Controlled Trial. *Frontiers in Psychiatry / Frontiers Research Foundation*, *9*. https://doi.org/10.3389/fpsyt.2018.00694

Gilburt, H., Rose, D., & Slade, M. (2008). The importance of relationships in mental health care: A qualitative study of service users' experiences of psychiatric hospital admission in the UK. *BMC Health Services Research*, *8*(1), 92. https://doi.org/10.1186/1472-6963-8-92

Hanssen, H., Minghetti, A., Faude, O., Schmidt-Trucksäss, A., Zahner, L., Beck, J., & Donath, L. (2018). Effects of different endurance exercise modalities on retinal vessel diameters in unipolar depression. *Microvascular Research*, *120*, 111–116. https://doi.org/10.1016/j.mvr.2018.07.003

Hanssen, H., Minghetti, A., & Faude, O., Schmidt-Trucksäss A, Zahner L, Beck J, Donath L. (2018). Effects of endurance exercise modalities on arterial stiffness in patients suffering from unipolar depression: A randomized controlled trial. *Frontiers in Psychiatry / Frontiers Research Foundation*. https://doi.org/10.3389/fpsyt.2017.00311

Haussleiter, I., Bolsinger, B., Assion, H. -J., & Juckel, G. (2020). Adjuvant Guided Exercise Therapy versus Self-Organized Activity in Patients with

Major Depression. *The Journal of Nervous and Mental Disease*, *208*(12), 982–988. https://doi.org/10.1097/NMD.0000000000001240

Heggelund, J., Nilsberg, G. E., Hoff, J., Morken, G., & Helgerud, J. (2011). Effects of high aerobic intensity training in patients with schizophrenia—A controlled trial. *Nordic Journal of Psychiatry*, *65*(4), 269–275. https://doi.org/10.3109/08039488.2011.560278

Heissel, A., Vesterling, A., & White, S. A., Kallies G, Behr D, Arafat AM, Reischies FM, Heinzel S, Budde H. (2015). Feasibility of an exercise program for older depressive inpatients: A pilot study. International Journal Of Geriatric Psychiatry, *28*(4), 163–171. https://doi.org/10.1024/1662-9647/a000134

Imboden, C., Gerber, M., & Beck, J., Eckert A, Lejri I, Pühse U, Holsboer-Trachsler E, Hatzinger M. (2021). Aerobic Exercise and Stretching as Add-On to Inpatient Treatment for Depression Have No Differential Effects on Sleep Measures. *Brain Sciences*, *11*, 411. https://doi.org/10.3390/brainsci11040411

Imboden, C., Gerber, M., Beck, J., Holsboer-Trachsler, E., Pühse, U., & Hatzinger, M. (2020). Aerobic exercise or stretching as add-on to inpatient treatment of depression: Similar antidepressant effects on depressive symptoms and larger effects on working memory for aerobic exercise alone. *Journal of Affective Disorders*, *276*, 866–876. https://doi.org/10.1016/j.jad.2020.07.052

Inskip, H. M., Harris, E. C., & Barraclough, B. (1998). Lifetime risk of suicide for affective disorder, alcoholism and schizophrenia. *The British Journal of Psychiatry: The Journal of Mental Science*, *172*(1), 35–37. https://doi.org/10.1192/bjp.172.1.35

Jae, K. H., Kil, S. B., So, B., Lee, O., Song, W., & Kim, Y. (2014). Increase of circulating BDNF levels and its relation to improvement of physical fitness following 12 weeks of combined exercise in chronic patients with schizophrenia: A pilot study. *Psychiatry Research*, *220*(3), 792–796. https://doi.org/10.1016/j.psychres.2014.09.020

Kahl, K. G., Kerling, A., Tegtbur, U., Gützlaff, E., Herrmann, J., Borchert, L., Ates, Z., Westhoff-Bleck, M., Hueper, K., & Hartung, D. (2016). Effects of additional exercise training on epicardial, intra-abdominal and subcutaneous adipose tissue in major depressive disorder: A randomized pilot study. *Journal of Affective Disorders*, *192*, 91–97. https://doi.org/10.1016/j.jad.2015.12.015

Kennedy, J. L., Altar, C. A., Taylor, D. L., Degtiar, I., & Hornberger, J. C. (2014). The social and economic burden of treatment-resistant schizophrenia. *International Clinical Psychopharmacology*, *29*(2), 63–76. https://doi.org/10.1097/yic.0b013e32836508e6

Kerling, A., Hartung, D., Stubbs, B., Kück, M., Tegtbur, U., Grams, L., Weber-Spickschen, T. S., & Kahl, K. G. (2018). Impact of aerobic exercise on muscle mass in patients with major depressive disorder: A randomized controlled trial. *Neuropsychiatric Disease and treatment*, *Volume 14*, 1969–1974. Published Online First: 2018. https://doi.org/10.2147/NDT.S167786

Kerling, A., Kück, M., Tegtbur, U., Grams, L., Weber-Spickschen, S., Hanke, A., Stubbs, B., & Kahl, K. G. (2017). Exercise increases serum brain-derived neurotrophic factor in patients with major depressive disorder. *Journal of Affective Disorders*, *215*, 152–155. https://doi.org/10.1016/j.jad.2017.03.034

Kerling, A., Tegtbur, U., & Gützlaff, E., Kück M, Borchert L, Ates Z, von Bohlen A, Frieling H, Hüper K, Hartung D, Schweiger U. (2015). Effects of adjunctive exercise on physiological and psychological parameters in depression: A randomized pilot trial. *Journal of Affective Disorders*, *177*, 1–6. https://doi.org/10.1016/j.jad.2015.01.006

Kerling, A., von Bohlen, A., Kück, M., Tegtbur, U., Grams, L., Haufe, S., Gützlaff, E., & Kahl, K. G. (2016). Exercise therapy improves aerobic capacity of inpatients with major depressive disorder. *Brain and Behavior*, *6*(6). https://doi.org/10.1002/brb3.469

Kern, L., Morvan, Y., Mattar, L., Molina, E., Tailhardat, L., Peguet, A., De Tournemire, R., Hirot, F., Rizk, M., Godart, N., & Fautrelle, L. Development and evaluation of an adapted physical activity program in anorexia nervosa inpatients: A pilot study. (2020). *European Eating Disorders Review: The Journal of the Eating Disorders Association*, *28*(6), 687–700. Published Online First: 2020. https://doi.org/10.1002/erv.2779

Kim, Y. S., Song, B. K., & Oh, J. S., Woo SS. (2014). Aerobic exercise improves gastrointestinal motility in psychiatric inpatients. *World Journal of Gastroenterology: WJG*, *20*(30), 10577–10584. https://doi.org/10.3748/wjg.v20.i30.10577

Knapen, J., Van de Vliet, P., Van Coppenolle, H., David, A., Peuskens, J., Knapen, K., & Pieters, G. (2003). The effectiveness of two psychomotor therapy programmes on physical firness and physical self-concept in nonpsychotic psychiatric patients: A randomized controlled trial. *Clinical Rehabilitation*, *17)6*(6), 637–647. https://doi.org/10.1191/0269215503cr659oa

Knapen, J., Van De Vliet, P., Van Coppenolle, H., David, A., Peuskens, J., Pieters, G., & Knapen, K. (2005). Comparison of changes in physical self-concept, global self-esteem, depression and anxiety following two different psychomotor therapy programs in nonpsychotic psychiatric inpatients. *Psychotherapy and Psychosomatics*, *74*(6), 353–361. https://doi.org/10.1159/000087782

Knubben, K., Reischies, F. M., Adli, M., Schlattmann, P., Bauer, M., Dimeo, F., & Ansley, L. (2007). A randomised, controlled study on the effects of a short-term endurance training programme in patients with major depression. *British Journal of Sports Medicine*, *41*(1), 29–33. https://doi.org/10.1136/bjsm.2006.030130

Korman, N., Armour, M., Chapman, J., Rosenbaum, S., Kisely, S., Suetani, S., Firth, J., & Siskind, D. (2020). High Intensity Interval training (HIIT) for people with severe mental illness: A systematic review & meta-analysis of intervention studies– considering diverse approaches for mental and physical recovery. *Psychiatry Research*, *284*, 112601. https://doi.org/10.1016/j.psychres.2019.112601

Kruisdijk, F., Deenik, J., Tenback, D., Tak, E., Beekman, A. -J., van Harten, P., Hopman-Rock, M., & Hendriksen, I. (2017). Accelerometer-measured sedentary behaviour and physical activity of inpatients with severe mental illness. *Psychiatry Research*, *254*, 67–74. https://doi.org/10.1016/j.psychres.2017.04.035

Kurebayashi, Y., Mori, K., & Otaki, J. (2022). Effects of mild-intensity physical exercise on neurocognition in inpatients with schizophrenia: A pilot randomized controlled trial. *Perspectives in Psychiatric Care*, *2021*(3), 1–11. https://doi.org/10.1111/ppc.12896

Lambert, T. J. R., Reavley, N. J., Jorm, A. F., & Oakley Browne, M. A. Royal Australian and New Zealand College of Psychiatrists expert consensus statement for the treatment, management and monitoring of the physical health of people with an enduring psychotic illness. (2017). *The Australian and New Zealand Journal of Psychiatry*, *51*(4), 322–337. Published Online First: 2017. https://doi.org/10.1177/0004867416686693

Lan, Y. L., Ping, L. Y., Su, L. W., & Chen, C. C. (2022). The Impact of Health Promotion Activities on the Physiological, Psychological, and Social Functions of Inpatients with Chronic Mental Illness. *Psychiatry Investigation*, *19*(3), 171–177. https://doi.org/10.30773/pi.2021.0128

Legrand, F. D., & Neff, E. M. (2016). Efficacy of exercise as an adjunct treatment for clinically depressed inpatients during the initial stages of antidepressant pharmacotherapy: An open randomized controlled trial. *Journal of Affective Disorders*, *191*, 139–144. https://doi.org/10.1016/j.jad.2015.11.047

Lohr, J. B., Palmer, B. W., Eidt, C. A., Aailaboyina, S., Mausbach, B. T., Wolkowitz, O. M., Thorp, S. R., & Jeste, D. V. (2015). Is post-traumatic stress disorder associated with premature senescence? A review of the literature. *The American Journal of Geriatric Psychiatry: Official Journal of the American Association for Geriatric Psychiatry*, *23*(7), 709–725. https://doi.org/10.1016/j.jagp.2015.04.001

Malhi, G. S., & Byrow, Y. 2016 Advance online publication. Exercising control over bipolar disorder. *Evidence-Based Mental Health*, *19*(4), 103–105, https://doi.org/10.1136/eb-2016-102430

Manjunath, R. B., Varambally, S., Thirthalli, J., Basavaraddi, I. V., & Gangadhar, B. N. (2013). Efficacy of yoga as an add-on treatment for in-patients with functional psychotic disorder. *Indian Journal of Psychiatry*, *55*(7), S374–378. https://doi.org/10.4103/0019-5545.116314

Martinsen, E. W., Hoffart, A., & Ø, S. (1989). Comparing aerobic with non-aerobic forms of exercise in the treatment of clinical depression: A randomized trial. *Comprehensive Psychiatry*, *30*(4), 324–331. https://doi.org/10.1016/0010-440X(89)90057-6

Martinsen, E. W., Hoffart, A., & Solberg, Y. (1989). Aerobic and non-aerobic forms of exercise in the treatment of anxiety disorders. *Stress Medicine*, *5*(2), 115–120. https://doi.org/10.1002/smi.2460050209

Martinsen, E. W., Sandvik, L., & Kolbjørnsrud, O. B. (1989). Aerobic exercise in the treatment of nonpsychotic mental disorders: An exploratory study. *Nordic Journal of Psychiatry*, *43*(6), 521–529. https://doi.org/10.3109/08039488909103250

Martland, R., Mondelli, V., Gaughran, F., & Stubbs, B. (2020). Can high intensity interval training improve health outcomes among people with mental illness? A systematic review and preliminary meta-analysis of intervention studies across a range of mental illnesses. *Journal of Affective Disorders*, *263*, 629–660. https://doi.org/10.1016/j.jad.2019.11.039

Mazyarkin, Z., Peleg, T., & Golani, I., Sharony L, Kremer I, Shamir A. (2019). Health benefits of a physical exercise program for inpatients with mental health; a pilot study. *Journal of Psychiatric Research*, *113*, 10–16. https://doi.org/10.1016/j.jpsychires.2019.03.002

McCartney, D., Isik, A. D., Rooney, K., Arnold, J. C., Bartlett, D. J., Murnion, B., Richards, E., Arkell, T. R., Lintzeris, N., & McGregor, I. S. (2021). The effect of daily aerobic cycling exercise on sleep quality during inpatient cannabis withdrawal: A randomised controlled trial. *Journal of Sleep Research*, *30*(3), 1–12. https://doi.org/10.1111/jsr.13211

Minghetti, A., Faude, O., & Hanssen, H., Zahner L, Gerber M, Donath L. (2018). Sprint interval training (SIT) substantially reduces depressive symptoms in major depressive disorder (MDD): A randomized controlled trial. *Psychiatry Research*, *265*, 292–297. https://doi.org/10.1016/j.psychres.2018.04.053

Morres, I. D., Hatzigeorgiadis, A., Stathi, A., Comoutos, N., Arpin-Cribbie, C., Krommidas, C., & Theodorakis, Y. (2019). Aerobic exercise for adult patients with major depressive disorder in mental health services: A systematic review and meta-analysis. *Depression and Anxiety*, *36*(1), 39–53. https://doi.org/10.1002/da.22842

National Institute for Health and Care Excellence (NICE). (2014). *Psychosis and schizophrenia in adults: Prevention and management*. https://www.nice.org.uk/guidance/cg178

Ng, F., Dodd, S., & Berk, M. (2007). The effects of physical activity in the acute treatment of bipolar disorder: A pilot study. *Journal of Affective Disorders*, *101*(1–3), 259–262. https://doi.org/10.1016/j.jad.2006.11.014

NICE guideline.Psychosis and schizophrenia in children and young people. 2013.

Palmer, J. A., Palmer, L. K., Michiels, K., & Thigpen, B. (1995). Effects of type of exercise on depression in recovering substance abusers. *Perceptual and Motor Skills*, *80*(2), 523–530. https://doi.org/10.2466/pms.1995.80.2.523doi:10.2466/pms.1995.80.2.523

Philippot, A., Dubois, V., Lambrechts, K., Grogna, D., Robert, A., Jonckheer, U., Chakib, W., Beine, A., Bleyenheuft, Y., & De Volder, A. G. (2022). Impact of physical exercise on depression and anxiety in adolescent inpatients: A randomized controlled trial. *Journal of Affective Disorders*, *301*, 145–153. https://doi.org/10.1016/j.jad.2022.01.011

Roest, A. M., Martens, E. J., de Jonge, P., & Denollet, J. (2010). Anxiety and Risk of Incident Coronary Heart Disease. A Meta-Analysis. *Journal of the American College of Cardiology*, *56*(1), 38–46. https://doi.org/10.1016/j.jacc.2010.03.034

Rosenbaum, S., Sherrington, C., & Tiedemann, A. Exercise augmentation compared with usual care for post-traumatic stress disorder: A randomized controlled trial. (2015). *Acta Psychiatrica Scandinavica*, *131*(5), 350–359. Published Online First: 2015. https://doi.org/10.1111/acps.12371

Roy, A., Govindan, R., & Muralidharan, K. (2018). The impact of an add-on video assisted structured aerobic exercise module on mood and somatic symptoms among women with depressive disorders: Study from a tertiary care centre in India. *Asian Journal of Psychiatry*, *32*, 118–122. https://doi.org/10.1016/j.ajp.2017.12.004

Salehi, I., Hosseini, S. M., Haghighi, M., Jahangard, L., Bajoghli, H., Gerber, M., Pühse, U., Holsboer-Trachsler, E., & Brand, S. (2016). Electroconvulsive therapy (ECT) and aerobic exercise training (AET) increased plasma BDNF and ameliorated depressive symptoms in patients suffering from major depressive disorder. *Journal of Psychiatric Research*, *76*, 1–8. https://doi.org/10.1016/j.jpsychires.2016.01.012

Schneider, B. C., Moritz, S., Hottenrott, B., Reimer, J., Andreou, C., & Jelinek, L. (2016). Association Splitting: A randomized controlled trial of a new method to reduce craving among inpatients with alcohol dependence. *Psychiatry Research*, *238*, 310–317. Published Online First: 2016. https://doi.org/10.1016/j.psychres.2016.02.051

Schuch, F. B., Vasconcelos-Moreno, M. P., & Borowsky, C., Zimmermann AB, Wollenhaupt-Aguiar B, Ferrari P, de Almeida Fleck MP. (2014). The effects of exercise on oxidative stress (TBARS) and BDNF in severely depressed inpatients. *European Archives of Psychiatry and Clinical Neuroscience*, *264*(7), 605–613. https://doi.org/10.1007/s00406-014-0489-5

Schuch, F. B., Vasconcelos-Moreno, M. P., Borowsky, C., Zimmermann, A. B., Rocha, N. S., & Fleck, M. P. (2015). Exercise and severe major depression: Effect on symptom severity and quality of life at discharge in an inpatient cohort. *Journal of Psychiatric Research*, *61*, 25–32. https://doi.org/10.1016/j.jpsychires.2014.11.005

Sexton, H., Mære, D., & Dahl, N. H. (1989). Exercise intensity and reduction in neurotic symptoms: A controlled follow-up study. *Acta Psychiatrica Scandinavica*, *80*(3), 231–235. https://doi.org/10.1111/j.1600-0447.1989.tb01332.x

Shachar-Malach, T., Cooper Kazaz, R., Constantini, N., Lerer B. (2015). Effectiveness of Aerobic Exercise as an Augmentation Therapy for Inpatients with Major Depressive Disorder: A Preliminary Randomized Controlled Trial. *The Israel Journal of Psychiatry and Related Sciences*, *52*(3), 65–70.

Shiers, D., & Curtis, J. (2014). Cardiometabolic health in young people with psychosis. *The Lancet Psychiatry*, *1*(7), 492–494. https://doi.org/10.1016/S2215-0366(14)00072-8

Shimada, T., Ito, S., Makabe, A., Yamanushi, A., Takenaka, A., Kawano, K., & Kobayashi, M. (2020). Aerobic exercise and cognitive functioning in schizophrenia: Results of a 1-year follow-up from a randomized controlled trial. *Psychiatry Research*, *286*, 112854. https://doi.org/10.1016/j.psychres.2020.112854

Shimada, T., Ito, S., & Makabe, A., Yamanushi A, Takenaka A, Kobayashi M (2019). Aerobic exercise and cognitive functioning in schizophrenia: A pilot randomized controlled trial. *Psychiatry Research*, *282*, 112638. https://doi.org/10.1016/j.psychres.2019.112638

Slade, S. C., Dionne, C. E., Underwood, M., & Buchbinder, R. (2016). Consensus on Exercise Reporting Template (CERT): Explanation and Elaboration Statement. *British Journal of Sports Medicine*, *50*(23), 1428–1437. https://doi.org/10.1136/bjsports-2016-096651

Smith, M. A., & Lynch, W. J. (2012). Exercise as a potential treatment for drug abuse: Evidence from preclinical studies. *Frontiers in Psychiatry / Frontiers Research Foundation*, *2*. https://doi.org/10.3389/fpsyt.2011.00082

Song, B. K., Kim, Y. S., Kim, H. S., Oh JW, Lee O, Kim JS. Combined exercise improves gastrointestinal motility in psychiatric in patients. *World J Clin Cases*. 2018. https://doi.org/10.12998/wjcc.v6.i8.207

Stanley, S. H., Ng, S. M., & Laugharne, J. D. E. (2019). The 'Fit for Life' exercise programme: Improving the physical health of people with a mental illness. *Psychol Heal Med*, *24*(2), 187–192. https://doi.org/10.1080/13548506.2018.1530366

Stanton, R., & Happell, B. (2014a). Exercise for mental illness: A systematic review of inpatient studies. *International Journal of Mental Health Nursing*, *23*(3), 232–242. https://doi.org/10.1111/inm.12045

Stanton, R., & Happell, B. (2014b). A systematic review of the aerobic exercise program variables for people with schizophrenia. *Current Sports Medicine Reports*, *13*(4), 260–266. https://doi.org/10.1249/JSR.0000000000000069

Sterne, J. A., Hernán, M. A., & Reeves, B. C., Savović J, Berkman ND, Viswanathan M, Henry D, Altman DG, Ansari MT, Boutron I, Carpenter JR. (2016). ROBINS-I: A tool for assessing risk of bias in non-randomised studies of interventions. *The BMJ*, *355*, 4–10. https://doi.org/10.1136/bmj.i4919

Sterne, J. A. C., Savović, J., & Page, M. J., Elbers RG, Blencowe NS, Boutron I, Cates CJ, Cheng HY, Corbett MS, Eldridge SM, Emberson JR. (2019). RoB 2: A revised tool for assessing risk of bias in randomised trials. *The BMJ*, *366*, 1–8. https://doi.org/10.1136/bmj.l4898

Stubbs, B., Rosenbaum, S., Vancampfort, D., Ward, P. B., & Schuch, F. B. (2016). Exercise improves cardiorespiratory fitness in people with depression: A meta-analysis of randomized control trials. *Journal of Affective Disorders*, *190*, 249–253. https://doi.org/10.1016/j.jad.2015.10.010

Stubbs, B., Vancampfort, D., Hallgren, M., Firth, J., Veronese, N., Solmi, M., Brand, S., Cordes, J., Malchow, B., Gerber, M., Schmitt, A., Correll, C. U., De Hert, M., Gaughran, F., Schneider, F., Kinnafick, F., Falkai, P., Möller, H. -J., & Kahl, K. G. (2018). EPA guidance on physical activity as a treatment for severe mental illness: A meta-review of the evidence and Position Statement from the

European Psychiatric Association (EPA), supported by the International Organization of Physical Therapists in Mental. *European Psychiatry*, *54*, 124–144. https://doi.org/10.1016/j.eurpsy.2018.07.004

SY, L., Abdullah, A., Abu Bakar, A. K., Thambu, M., & Nik Jaafar, N. R. (2016). Structured Walking and Chronic Institutionalized Schizophrenia Inmates: A pilot RCT Study on Quality of Life. *Global Journal of Health Science*, *8*(1), 238–248. https://doi.org/10.5539/gjhs.v8n1p238

Tsukue, I., & Shohoji, T. Movement therapy for alcoholic patients. (1981). *Journal of Studies on Alcohol*, *42*(1), 144–149. Published Online First: 1981. https://doi.org/10.15288/jsa.1981.42.144

Vancampfort, D., Rosenbaum, S., & Probst, M., Soundy A, Mitchell AJ, De Hert M, Stubbs B. (2015). Promotion of cardiorespiratory fitness in schizophrenia: A clinical overview and meta-analysis. *Acta Psychiatrica Scandinavica*, *132*(2), 131–143. https://doi.org/10.1111/acps.12407

Vancampfort, D., Stubbs, B., Venigalla, S. K., & Probst, M. (2015). Adopting and maintaining physical activity behaviours in people with severe mental illness: The importance of autonomous motivation. *Prev Med (Baltim)*, *81*, 216–220. https://doi.org/10.1016/j.ypmed.2015.09.006

Vogelzangs, N., Seldenrijk, A., Beekman, A. T. F., van Hout, H. P. J., de Jonge, P., & Penninx, B. W. J. H. (2010). Cardiovascular disease in persons with depressive and anxiety disorders. *Journal of Affective Disorders*, *125*(1–3), 241–248. https://doi.org/10.1016/j.jad.2010.02.112

Weinmann, S., Janssen, B., & Gaebel, W. (2004). Switching antipsychotics in inpatient schizophrenia care: Predictors and outcomes. *The Journal of Clinical Psychiatry*, *65*(8), 1099–1105. https://doi.org/10.4088/JCP.v65n0812

Weitoft, G. R., & Rosén, M. (2005). Is perceived nervousness and anxiety a predictor of premature mortality and severe morbidity? A longitudinal follow up of the Swedish survey of living conditions. *Journal of Epidemiology and Community Health*, *59*, 794–798. https://doi.org/10.1136/jech.2005.033076

Wheeler, A. J., Roennfeldt, H., Slattery, M., Krinks, R., & Stewart, V. (2018). Codesigned recommendations for increasing engagement in structured physical activity for people with serious mental health problems in Australia. *Heal Soc Care Community*, *26*(6), 860–870. https://doi.org/10.1111/hsc.12597

Whiteford, H. A., Degenhardt, L., Rehm, J., Baxter, A. J., Ferrari, A. J., Erskine, H. E., Charlson, F. J., Norman, R. E., Flaxman, A. D., Johns, N., Burstein, R., Murray, C. J., & Vos, T. (2013). Global burden of disease attributable to mental and substance use disorders: Findings from the Global Burden of Disease Study 2010. *The Lancet*, *382*(9904), 1575–1586. https://doi.org/10.1016/S0140-6736(13)61611-6

Woodward, M. L., Gicas, K. M., Warburton, D. E., White, R. F., Rauscher, A., Leonova, O., Su, W., Smith, G. N., Thornton, A. E., Vertinsky, A. T., Phillips, A. A., Goghari, V. M., Honer, W. G., & Lang, D. J. (2018). Hippocampal volume and vasculature before and after exercise in treatment-resistant schizophrenia. *Schizophrenia Research*, *202*, 158–165. https://doi.org/10.1016/j.schres.2018.06.054

Wright, K., Armstrong, T., Taylor, A., & Dean, S. 'It's a double edged sword': A qualitative analysis of the experiences of exercise amongst people with Bipolar Disorder. (2012). *Journal of Affective Disorders*, *136*(3), 634–642. Published Online First: 2012. https://doi.org/10.1016/j.jad.2011.10.017

Wunram, H. L., Hamacher, S., Hellmich, M., Volk, M., Jänicke, F., Reinhard, F., Bloch, W., Zimmer, P., Graf, C., Schönau, E., Lehmkuhl, G., Bender, S., & Fricke, O. (2018). Whole body vibration added to treatment as usual is effective in adolescents with depression: A partly randomized, three-armed clinical trial in inpatients. *European Child & Adolescent Psychiatry*, *27*(5), 645–662. https://doi.org/10.1007/s00787-017-1071-2

Wunram, H. L., Oberste, M., Ziemendorff, A., Hamacher, S., Kapanci, T., Heller, R., Blick, S., Bloch, W., Clajus, T. C., Schönau, E., Bender, S., & Fricke, O. (2021). Differential effects of ergometer-cycling and Whole-Body-Vibration training on serological BDNF and IGF-1 in the treatment of adolescent depression - is there an impact of BDNFp.Val66Met variants? *Physiology and Behavior*, *241*, 113596. https://doi.org/10.1016/j.physbeh.2021.113596